MANET C4ISTAR PLANNING

NEED

MANET C4ISTAR planning ensures that military operations can leverage the capabilities of mobile ad-hoc networks effectively, ensuring reliable communication, situational awareness, and operational success in dynamic and challenging environment.

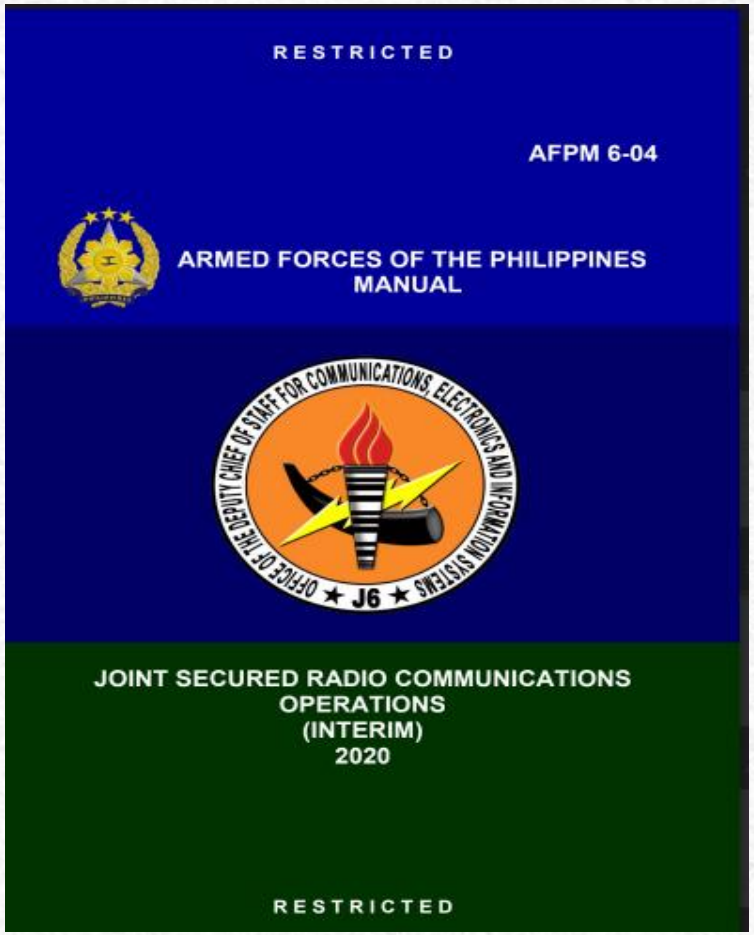
TIME

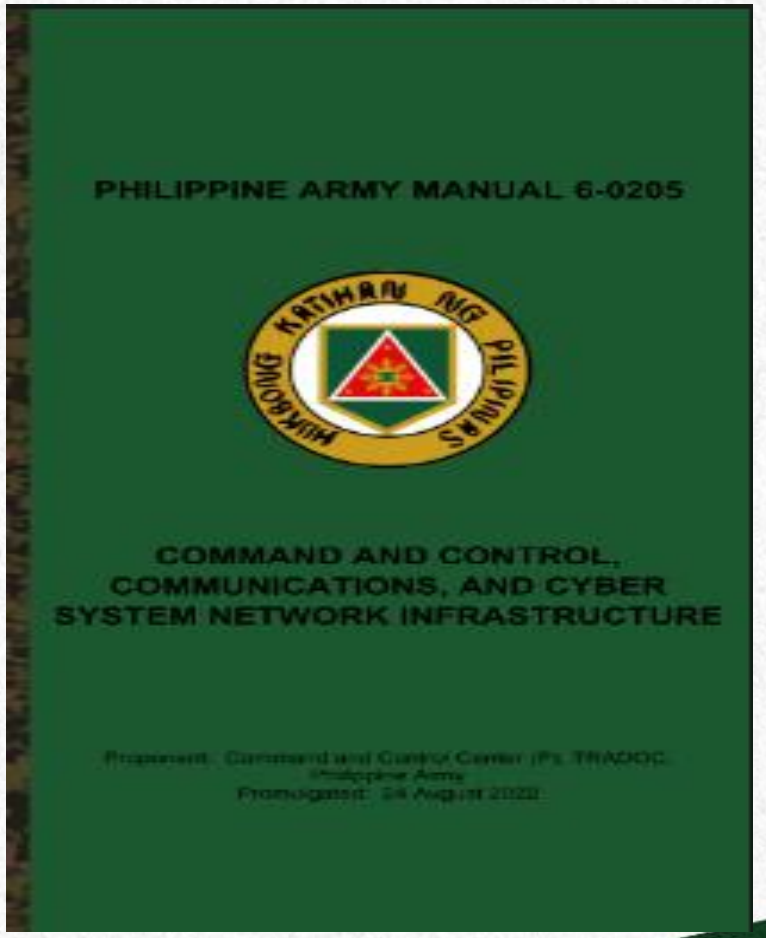


ROADMAP

- **Diagnose**
  - ✓Mobile Ad-hoc Networks(MANETs) capabilities in military operations.
  - ✓Importance of MANETs in enhancing situational awareness.
  - ✓Key components of MANET C4ISTAR Systems
  - ✓Technologies and standards used in MANETs.
  - ✓(C4ISTAR) systems in military operations.
  - ✓MANETs integration with C4ISTAR Systems.
  - ✓Different operational scenarios on MANET deployment strategies.
  - ✓Factors influencing MANET performance.
  - ✓Security measures and protocols to protect MANET communications.
  - ✓Strategies for enhancing MANET.
- **Experiment:** Ask them questions, recalls (Check on learning)
- **Reflect:** Let the students summarize the lesson
- **Evaluate:** Assess through Group Presentation (Case Scenario)

REFERENCES


RESTRICTED
AFPM 6-04
ARMED FORCES OF THE PHILIPPINES MANUAL
OFFICE OF THE DEPUTY CHIEF OF STAFF FOR COMMUNICATIONS, ELECTRONICS AND INFORMATION SYSTEMS
J6
JOINT SECURED RADIO COMMUNICATIONS OPERATIONS
(INTERIM)
2020
RESTRICTED


PHILIPPINE ARMY MANUAL 6-0205
KATIHAN NG PILIPINO
COMMAND AND CONTROL, COMMUNICATIONS, AND CYBER SYSTEM NETWORK INFRASTRUCTURE

LEARNING OBJECTIVE

At the end of the subject, the students will be able to demonstrate proficiency in planning and implementing MANET C4ISTAR systems to support military operations effectively in IAW PAM 6 00. (C3)

MANET C4ISTAR Planning

MANET C4ISTAR planning ensures that military operations can leverage the capabilities of mobile ad-hoc networks effectively, ensuring reliable communication, situational awareness, and operational success in dynamic and challenging environment.

MANET C4ISTAR Planning

E-LynX™ Networking the Battlefield







MANET C4ISTAR Planning

Modern Communication Challenges

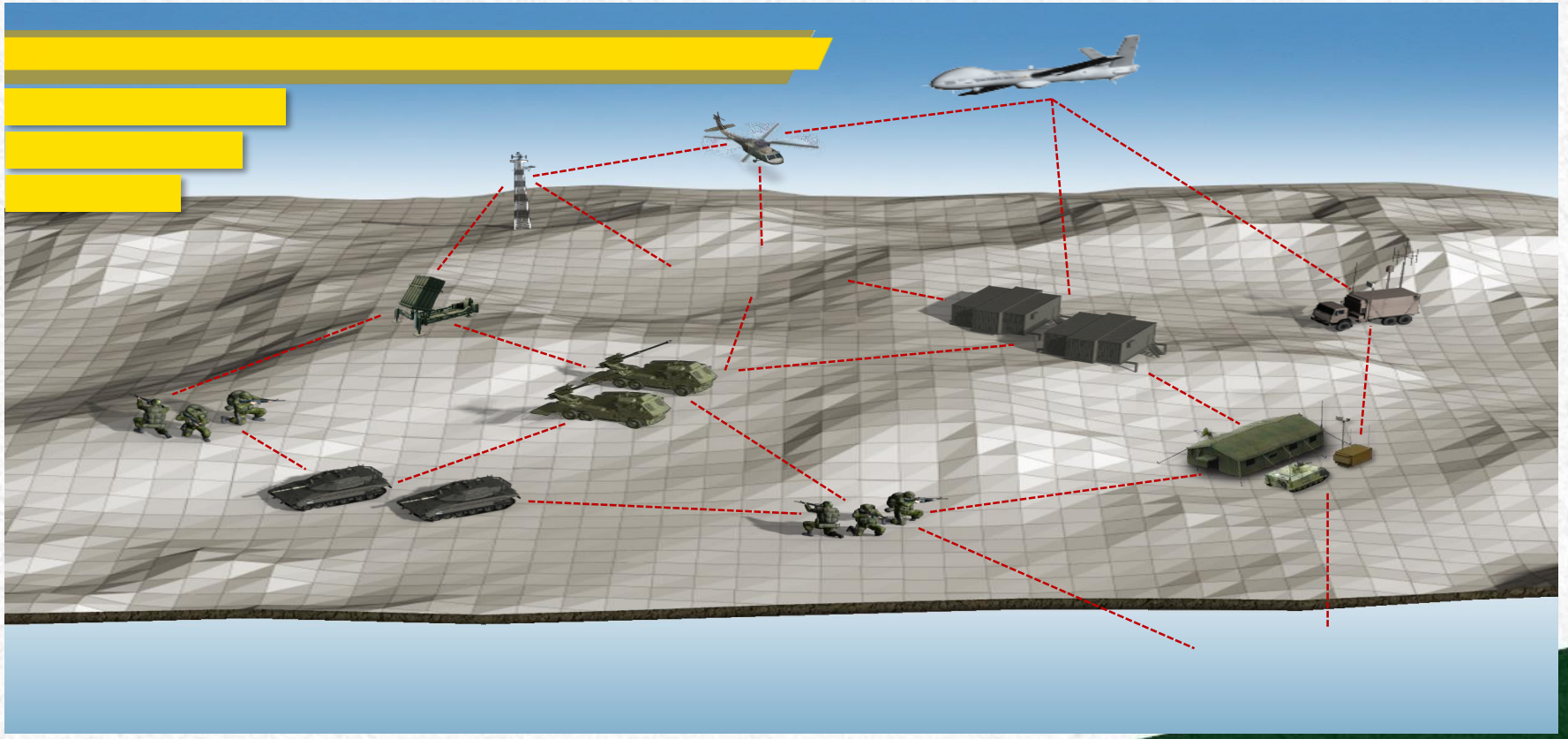
Modern tactical communication faces many challenges:

- Increase in capabilities:
  - ✓Higher data capacity and number of users per network
  - ✓More voice groups per Hertz, while:
- Adapting to harsh link conditions
- Dealing with Electronic Warfare
- Facing interferences from other transmitters



MANET C4ISTAR Planning

Modern Communication Challenges

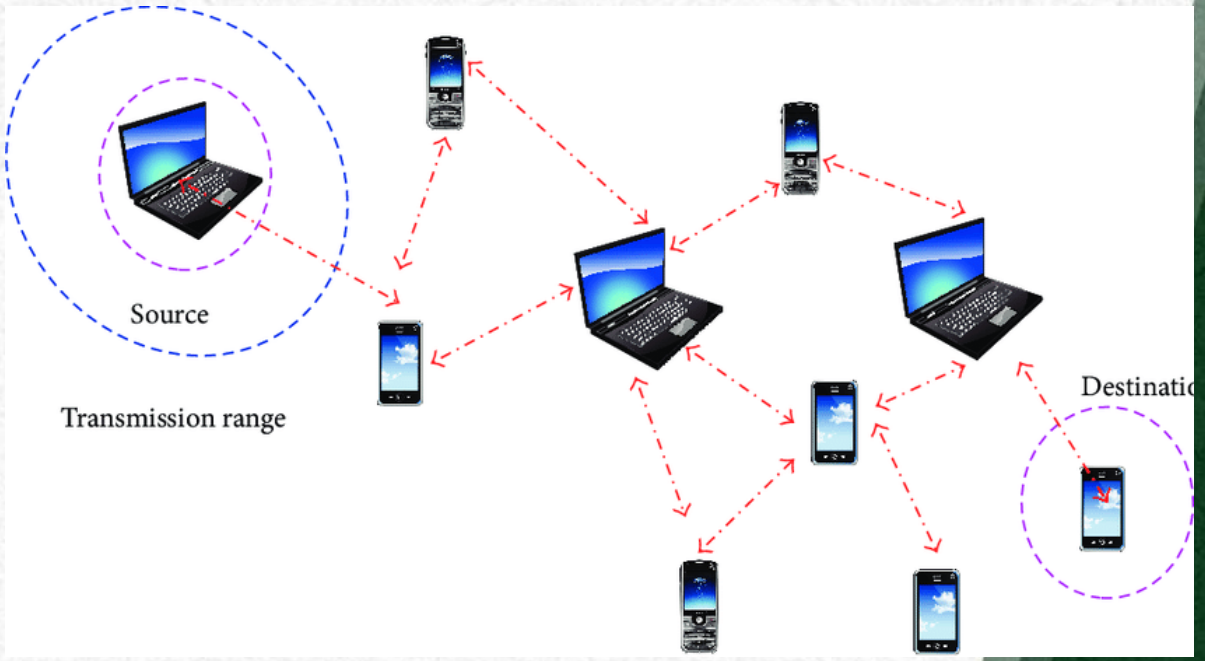


MANET C4ISTAR Planning

(MANET)

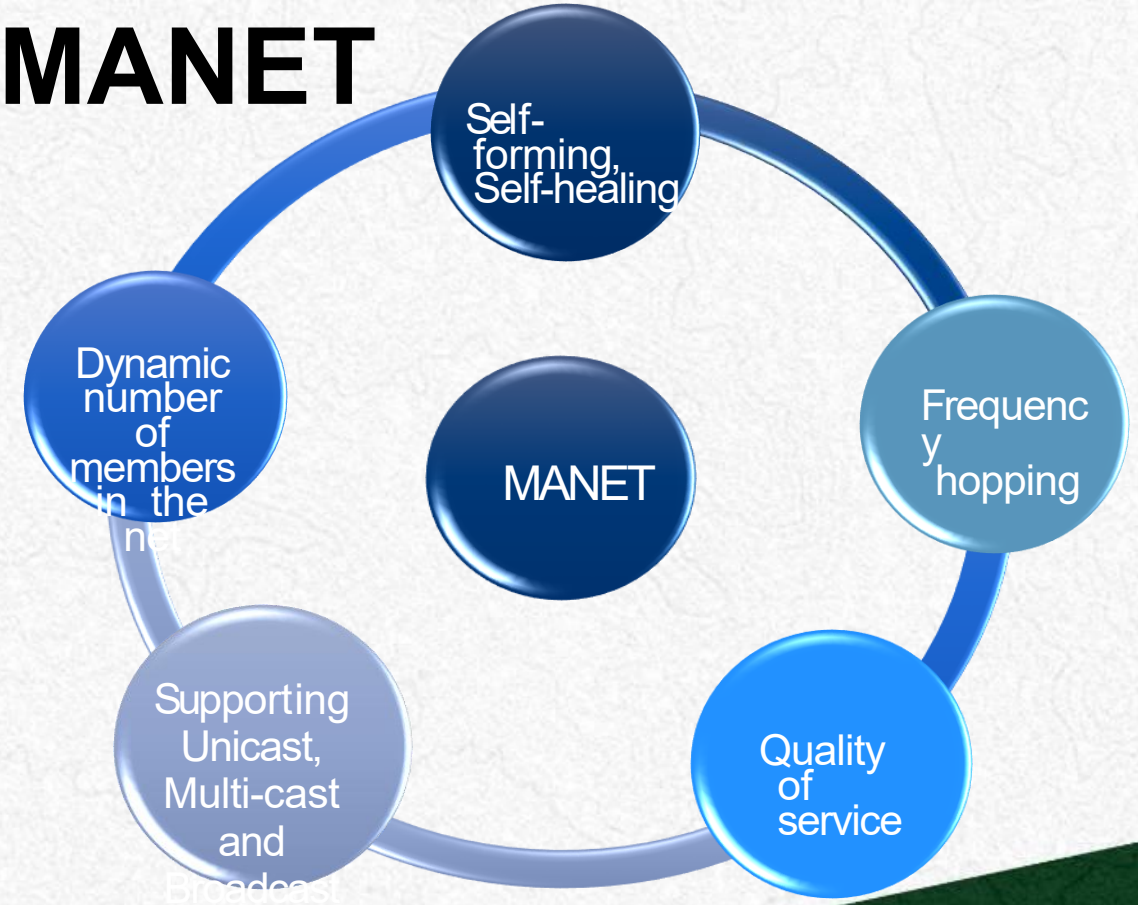
- A **MANET**, which stands for **mobile ad hoc network**, is a continuously self-configuring, infrastructure-less network of mobile devices connected without wires.

- It is defined as a collection of low-power wireless mobile nodes forming a temporary wireless network without the aid of any established infrastructure or centralized administration.



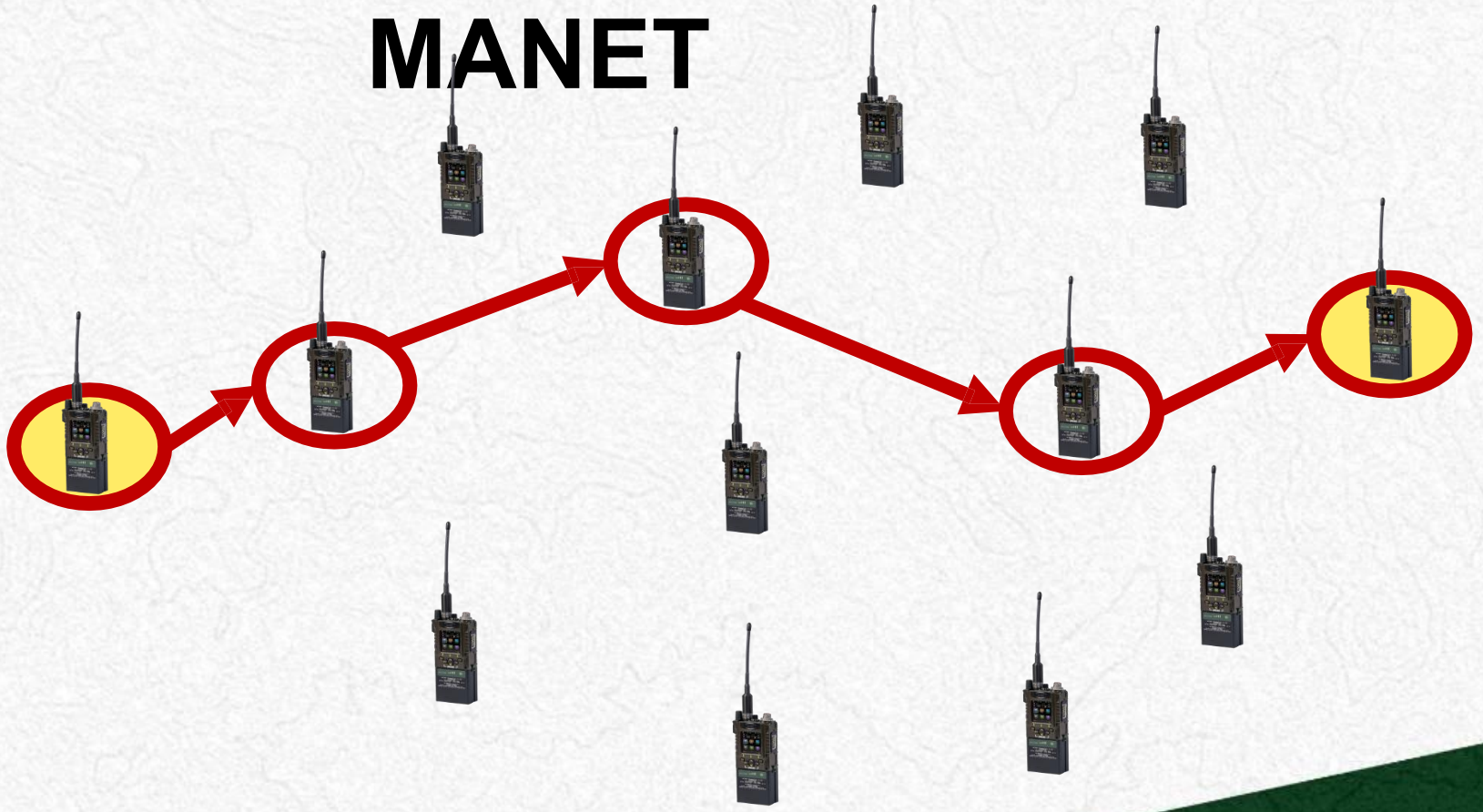
MANET C4ISTAR Planning

Main Features: MANET



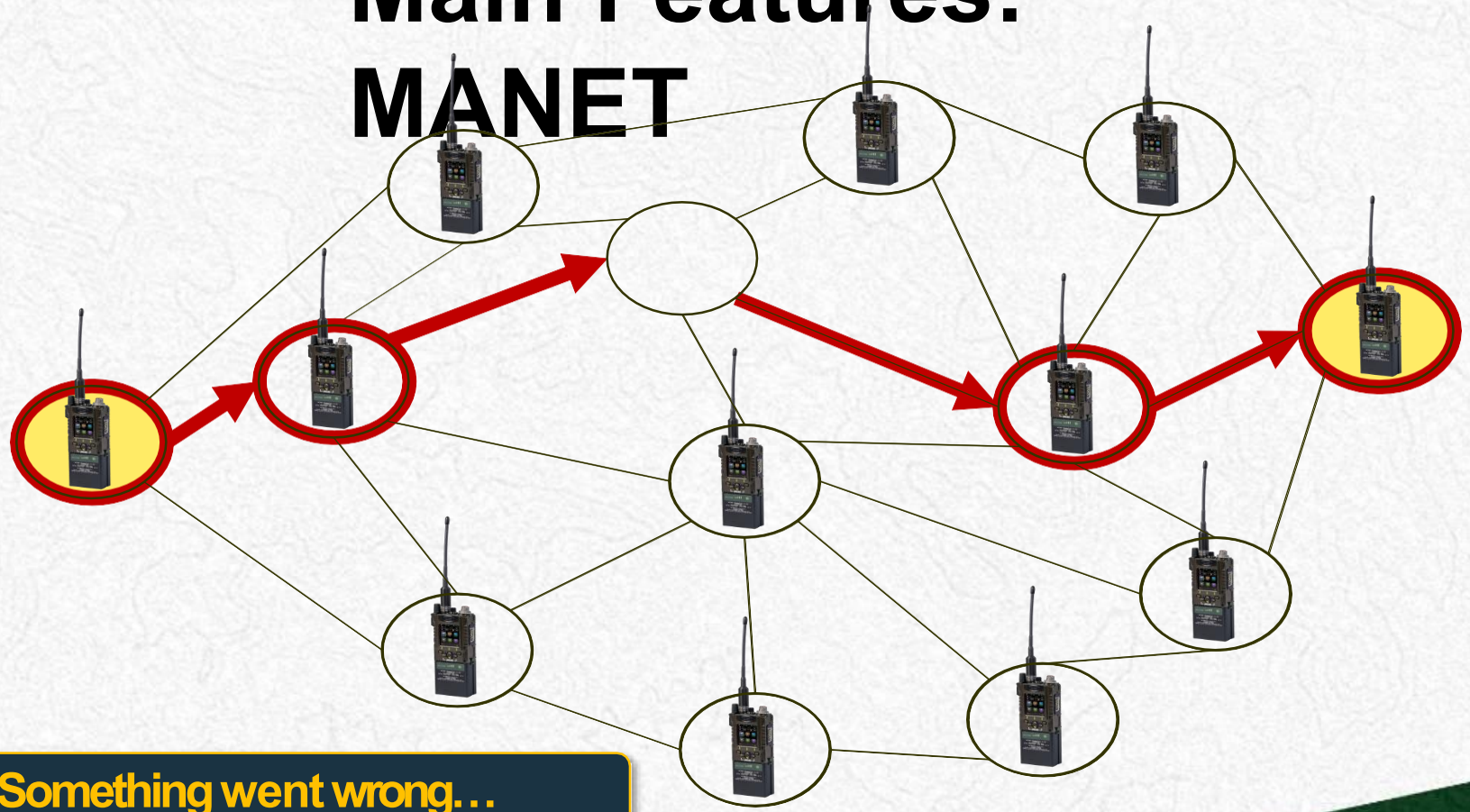
MANET C4ISTAR Planning

Main Features:
MANET



MANET C4ISTAR Planning

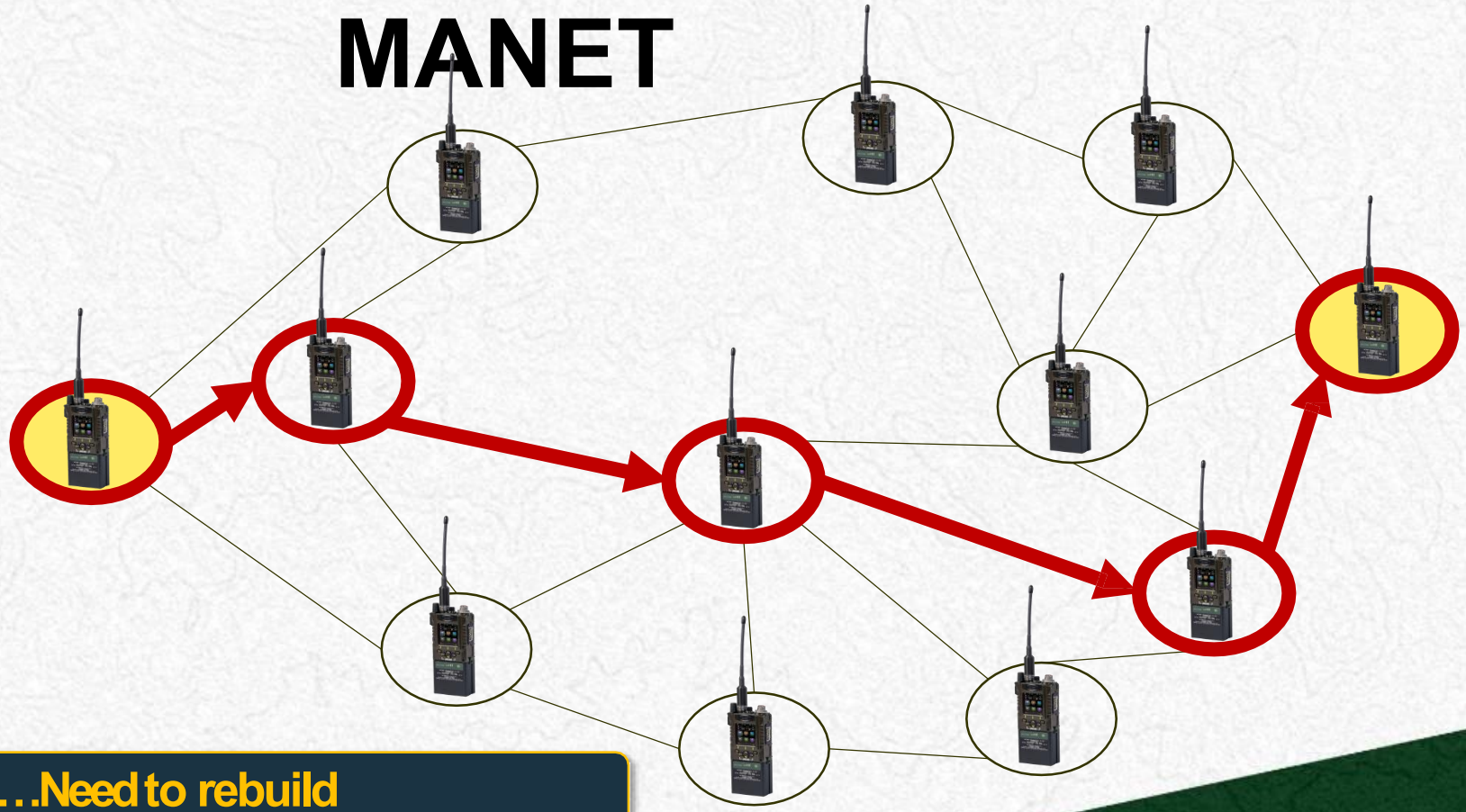
Main Features: MANET



Something went wrong…

MANET C4ISTAR Planning

Main Features:
MANET



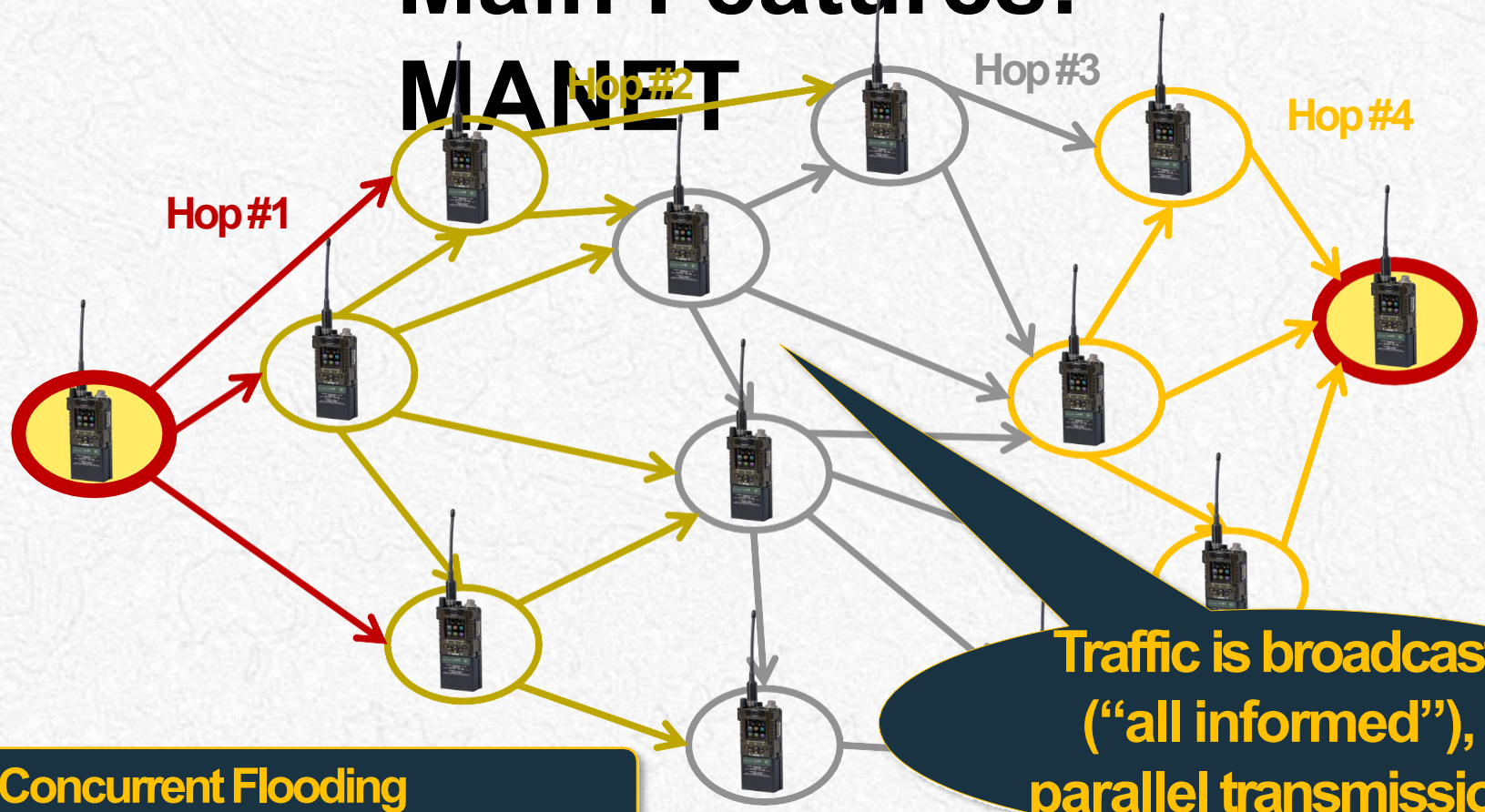
…Need to rebuild




Back In 5 Minutes

MANET C4ISTAR Planning

Main Features: MANET

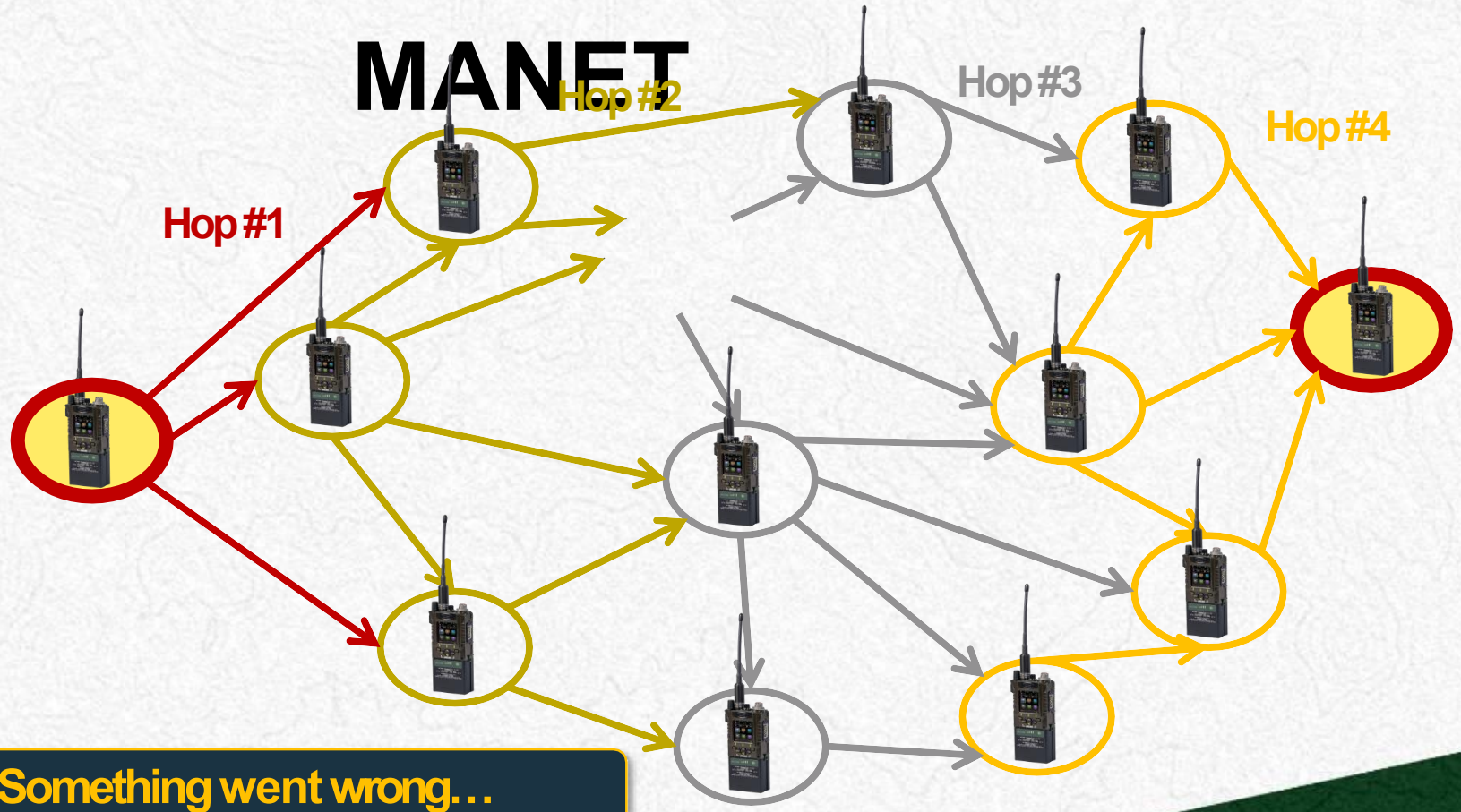


Concurrent Flooding

Traffic is broadcast ("all informed"), parallel transmission

MANET C4ISTAR Planning

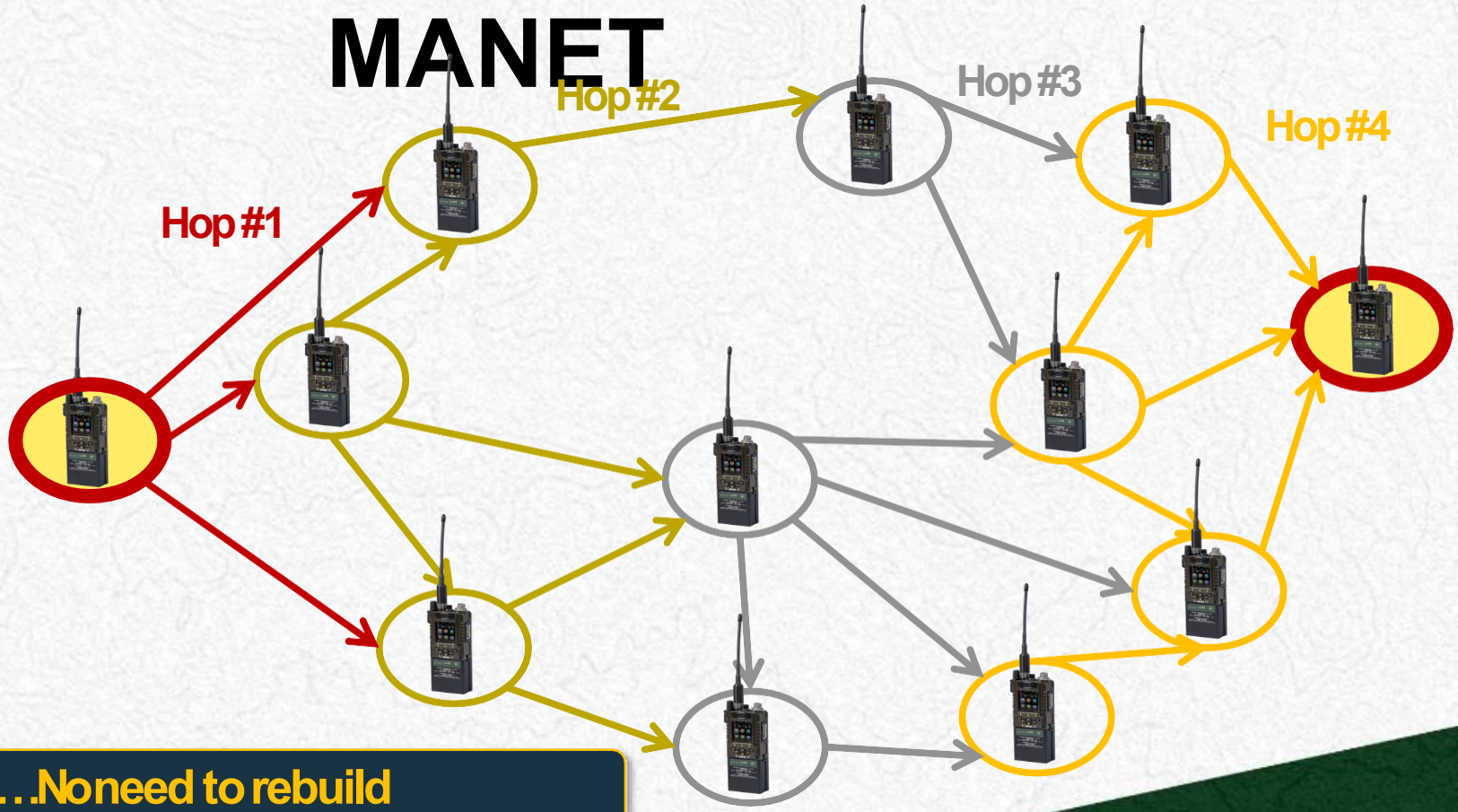
Main Features: MANET



Something went wrong…

MANET C4ISTAR Planning

Main Features: MANET



…Noneed to rebuild

MANET C4ISTAR Planning

Based on E-LynX™ SDR (Software Defined Radio) radio family, designed to meet the requirements for tactical communication

The E-LynX™ family is designed to be used in extreme combat conditions and allows reliable voice/data communication while making operation simple







MANET C4ISTAR Planning

E-LynX Setup

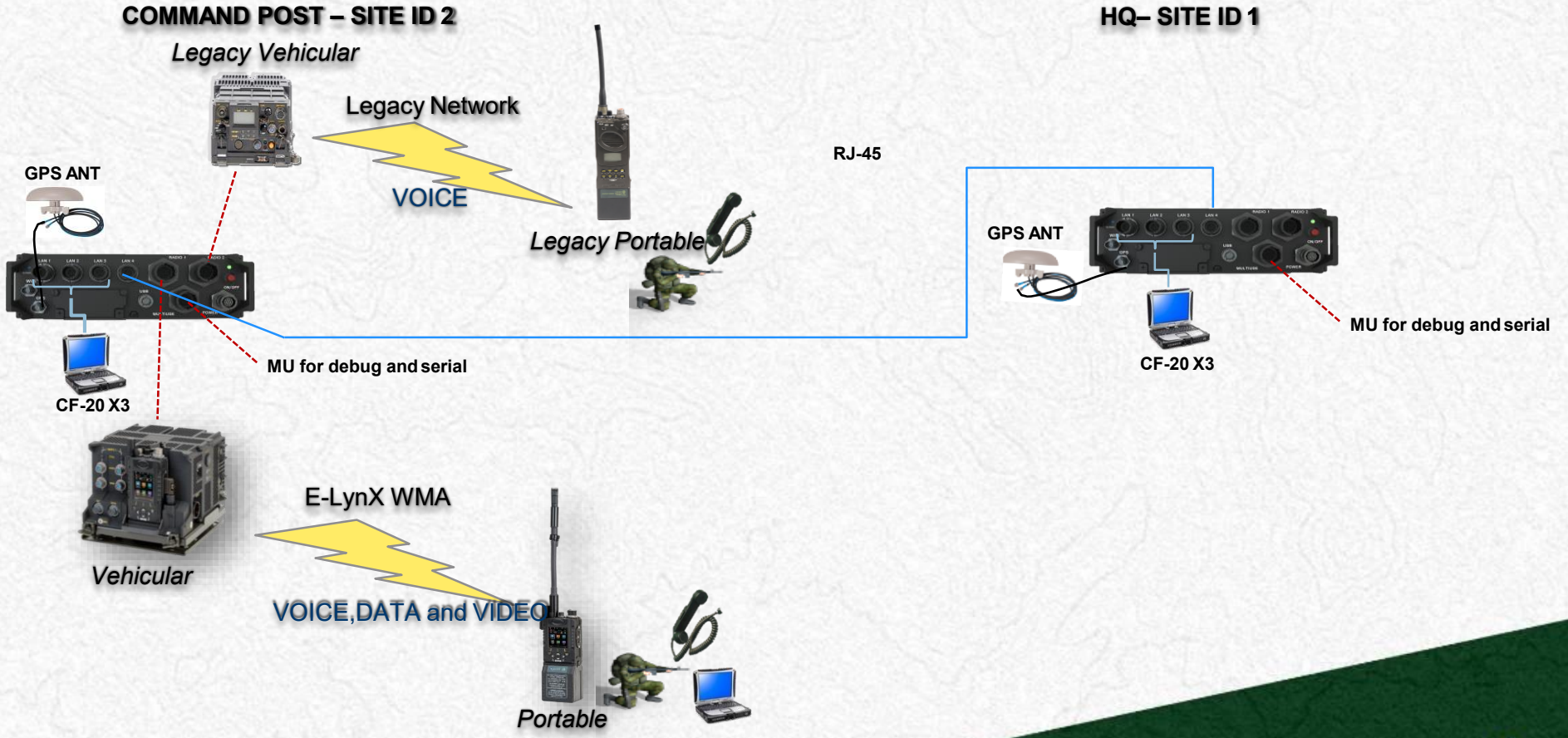






MANET C4ISTAR Planning

Command Post and HQ setup



MANET C4ISTAR Planning

E-LynX Preventive Maintenance



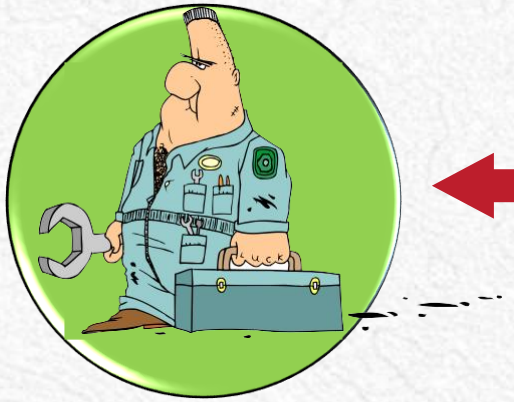




MANET C4ISTAR Planning

E-LynX Preventive Maintenance

Visual Inspection



Cleaning

MANET C4ISTAR Planning

General Safety Precautions

- Use only the designated tools (or the recommended tools) in order to perform maintenance tasks
- When needed, use protecting gloves

Preventive Maintenance Visual Inspection



MANET C4ISTAR Planning

- Before performing any maintenance actions, clean the equipment in accordance with the following procedure:

Inspect the exterior of the components.

Remove dust and other loose dirt with a soft clean cloth

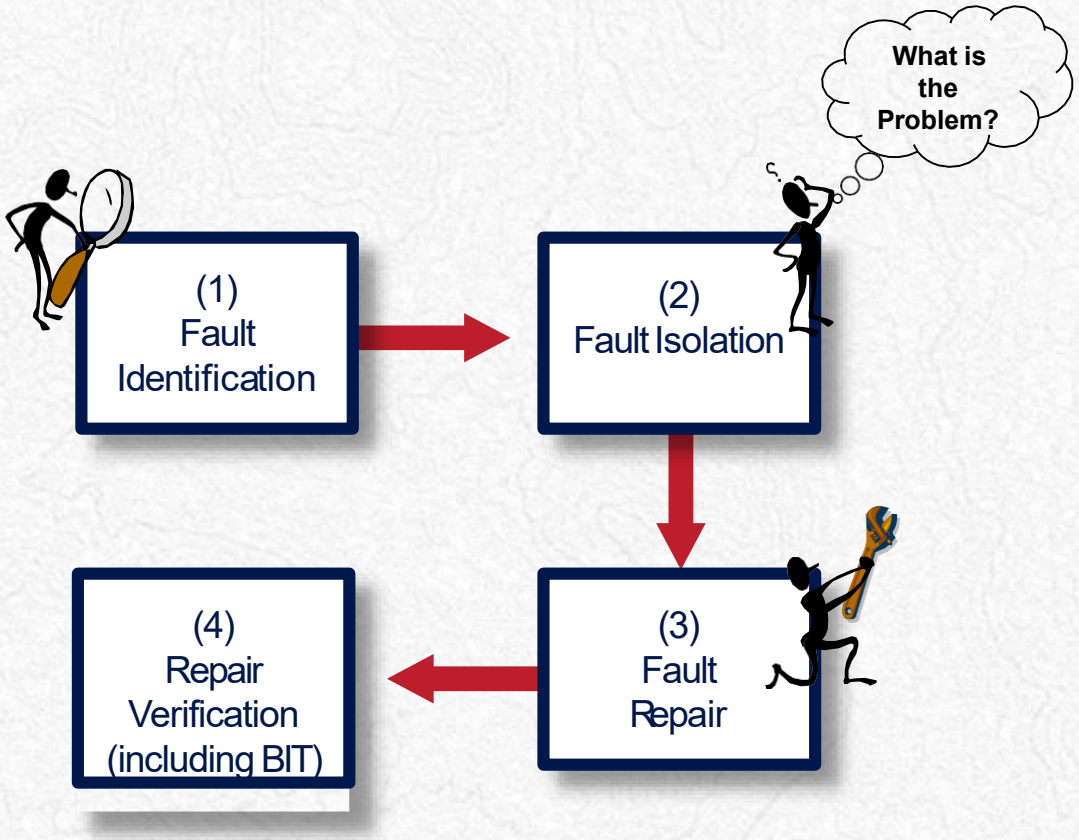


Clean the panel and controls using a soft, clean cloth.
If dirt is difficult to remove, dampen the cloth with mild soap and water prior to use.

Connecting the Dots

MANET C4ISTAR Planning

- Consists of 4 steps:



MANET Features and Capabilities

E-LynX Capabilities and Characteristics

- Simultaneous Voice, Data and Blue Force Tracking services via Unique Narrow and Wide band waveforms
- User friendly User Interface with Color Screen
- Preset configurations for versatile operational scenarios
- Network Management Tool for advanced radio parameters
- Built-In self Test (BIT) and diagnostics

MANET Features and Capabilities

- Simplified and unified operation and user interface for all set of radios taking into consideration usage by military forces
- Friendly intuitive user interface
- Same “look and feel” across all platforms
  - Different orientation between Hand-Held and Man-pack
  - Smartphone-like – very intuitive operation
- Local and Remote operation
  - Applies on all radio platforms and configurations
    - Hand-Held, Man-pack, Vehicular
  - Local operation
    - 2.8” Color display
    - 240 x 320 Pixels
    - Keypads
  - Remote operation
    - Replicates the local operation
    - IP/Ethernet based





MANET Features and Capabilities

- All E-Lynx™ configurations incorporate built-in information security capabilities
  - Embedded TRANSEC (Transmission Security) based on AES 256
  - Features anti-tamper and erase abilities
  - Keys management and loading
- Over the Air security is based on
  - Frequency Hopping (FH), TRANSEC
- Management security is based on SNMP(v3)
  - Supporting both Confidentiality and Authentication of data exchanged between the radio and the management station

MANET Features and Capabilities

Waveforms

- Narrowband Waveform (NB)
- Wideband Waveform (WB)
- Wideband Mobile Ad-hoc (WMA) - High Capacity (HC) waveform
- STANAG 4204 (S4204)
- CNR

MANET Features and Capabilities

Waveforms – WB Highlights

- 1 MHZ channel bandwidth
- Supports up to 4 concurrent voice sessions
- Analog and IP based voice services
- Multiple RF hop (static \ dynamic) TDMA based IP
- Network (Unicast and Multicast)
- TRANSEC protected, jamming resistant, Frequency hopping
- No single point of failure MANET, Global Navigation Satellite System (GNSS) independent
- Up to 140 nodes in a single network

MANET Features and Capabilities

Waveforms – WB Highlights

- Channel bandwidth of 25 \ 50 kHz (VHF and UHF)
- Up to 2 \ 4 simultaneous voice groups
- Analog and IP based voice services
- Up to 120 BFT sources in a single network
- Multiple RF hop (static), TDMA based IP network (Unicast and Multicast)
- TRANSEC protected, jamming resistant, Frequency hopping
- No single point of failure MANET, Global Navigation Satellite System (GNSS) independent
- Up to 60 nodes in a single network

0 nodes in a single network

MANET Features and Capabilities

Waveforms – HC/WMA Highlights

- 1 \ 2 \ 4 MHZ channel bandwidth
- Supports 7 concurrent voice sessions
- Analog and IP based voice services
- Automated self-forming, self-healing IP network
- Support Unicast and Multicast IP traffic
- TRANSEC protected, jamming resistant, Frequency hopping
- No single point of failure MANET, GNSS independent
- Up to 250 nodes in a single network

0 nodes in a single network

MANET Features and Capabilities

Waveforms – STANAG 4204 Highlights

- Compatibility with NATO waveform
- Non-encrypted
- Analog voice session

0 nodes in a single network

MANET Features and Capabilities

Waveforms – CNR Highlights

- Compatibility with Elbit legacy systems
- Encrypted
- Analog & Digital voice session




Back In 5 Minutes

MANET Features and Capabilities

Parts & Functions Elynx MCTR 7200HH

- Single channel radio
- 5W output power
- Size (mm): 137x75x53
- Deliverables:
  - 1 Hand held soldier radio
    - Including GPS antenna
  - 1 handset

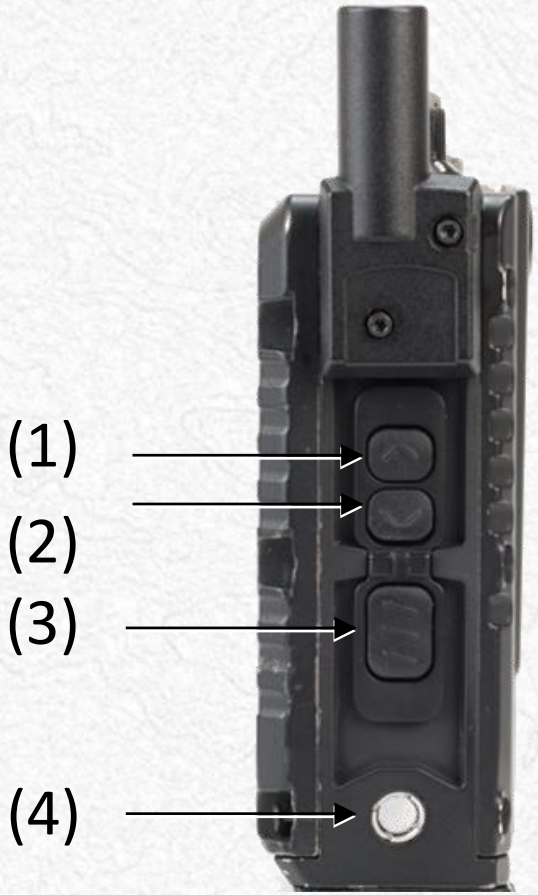


Parts & Functions Elynx MCTR 7200HH



No.	Item	Description
1	RF Antenna Connector	**RF antenna connector**
2.	Channel Selector	**The channel selector enables selecting between channels 0 through 9**
3.	Audio Connector	**Analog audio connection for handset**
4.	Speaker	**Speaker**
5.	Data Connector	**22 Pin data connector for external host, G-10N Key Fill device for loading encryption keys.**
6.	Screen	**Displays the GUI (Graphic User Interface)**
7.	Microphone	**Microphone**
8.	Battery Clamp	**Used to lock and release battery**
9.	Keypad	**Six-keys keypad used to control the GUI (Graphic User Interface)**
10.	GPS Antenna	**GPS antenna**

Parts & Functions Elynx MCTR 7200HH



No.	Item	Description
1.	Volume Up button	**Volume buttons, increases the volume level in the audio accessory**
2.	Volume Down button	**Volume buttons, decreases the volume level in the audio accessory**
3.	Push To Talk (PTT) button	**PTT button, enables transmitting without an external audio accessory**
4.	Breather	**Adjusts the internal pressure**

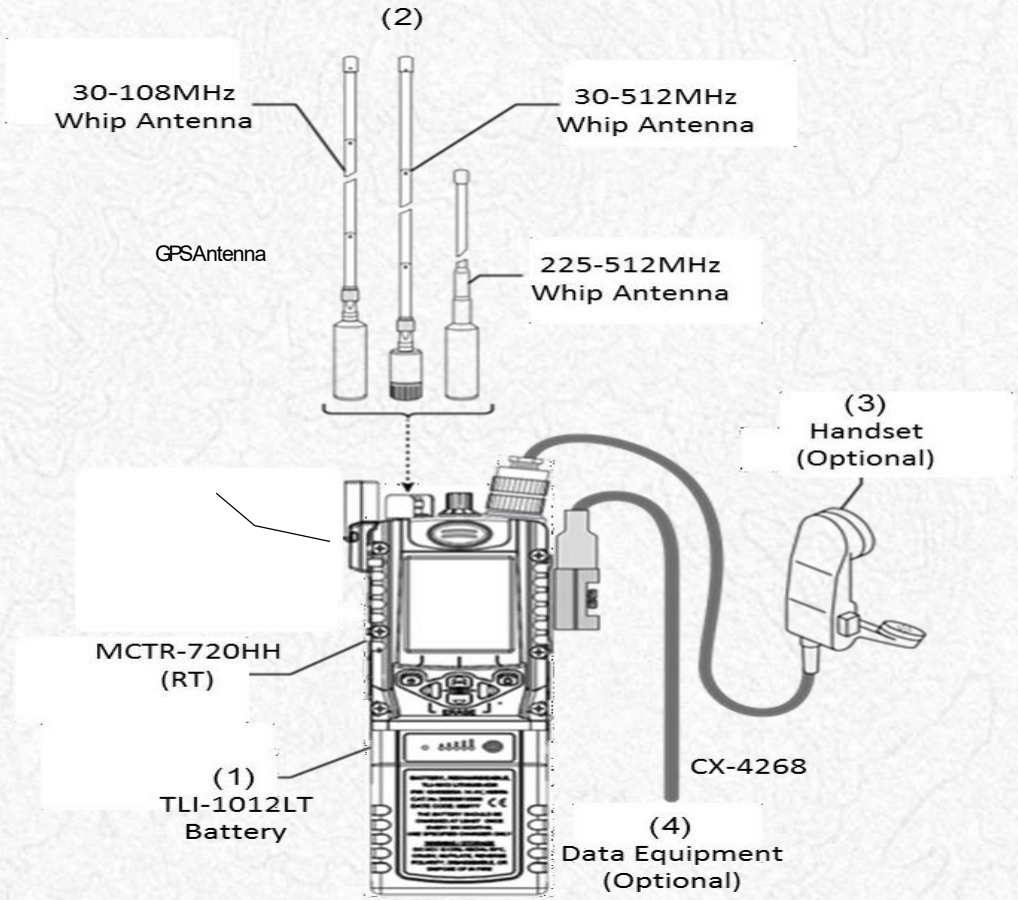
Parts & Functions Elynx MCTR 7200HH

- The TRANSEC and administrative keys are stored in the backed- up SRAM (Static Random Access Memory).
- Pressing left and right navigation keys together for 3 seconds erases the content SRAM.
- After keys erasure, the radio can only operate in Clear STANAG 4204 waveform.
- In order to resume full operation for all other waveforms, new TRANSEC keys and a configuration file consisting of administrative keys must be loaded.



Parts & Functions Elynx MCTR 7200HH

- E-LynX – HH

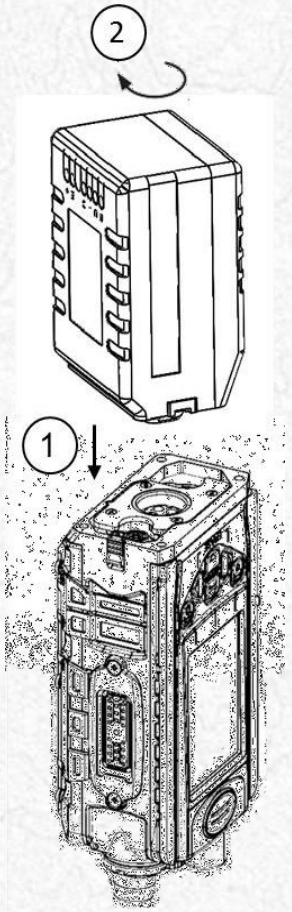


Parts & Functions Elynx MCTR 7200HH

- E-LynX – HH

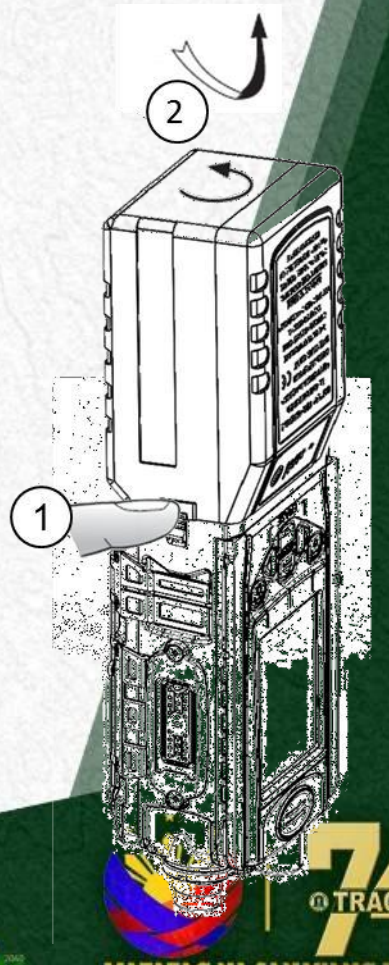
- Inserting the battery to the radio

1. Insert the battery facing the bottom part of the HH radio.
2. Press and rotate the battery clockwise to tighten.



- Removing the battery from the radio

1. Slide the battery catch.
2. Rotate the battery counter-clockwise to loosen.



Parts & Functions Elynx MCTR 7200HH

- An external handset and speaker can be connected to the standard 5-pin audio connector
- Handset H-250
  - Enables sending PTT commands for voice transmission
  - Connects to the E-LynX radio via an audio connector



Parts & Functions Elynx MCTR 7200HH

- All radio devices are loaded with Keys via the G-10N key fill device



Parts & Functions Elynx MCTR 7200HH VS50-VA

- Single channel vehicular adapter
- 50W output power
- Size (mm): 160x230x250
- Deliverables:
  - Single 50W transceiver
  - Mechanical mount
  - Set of mating connectors

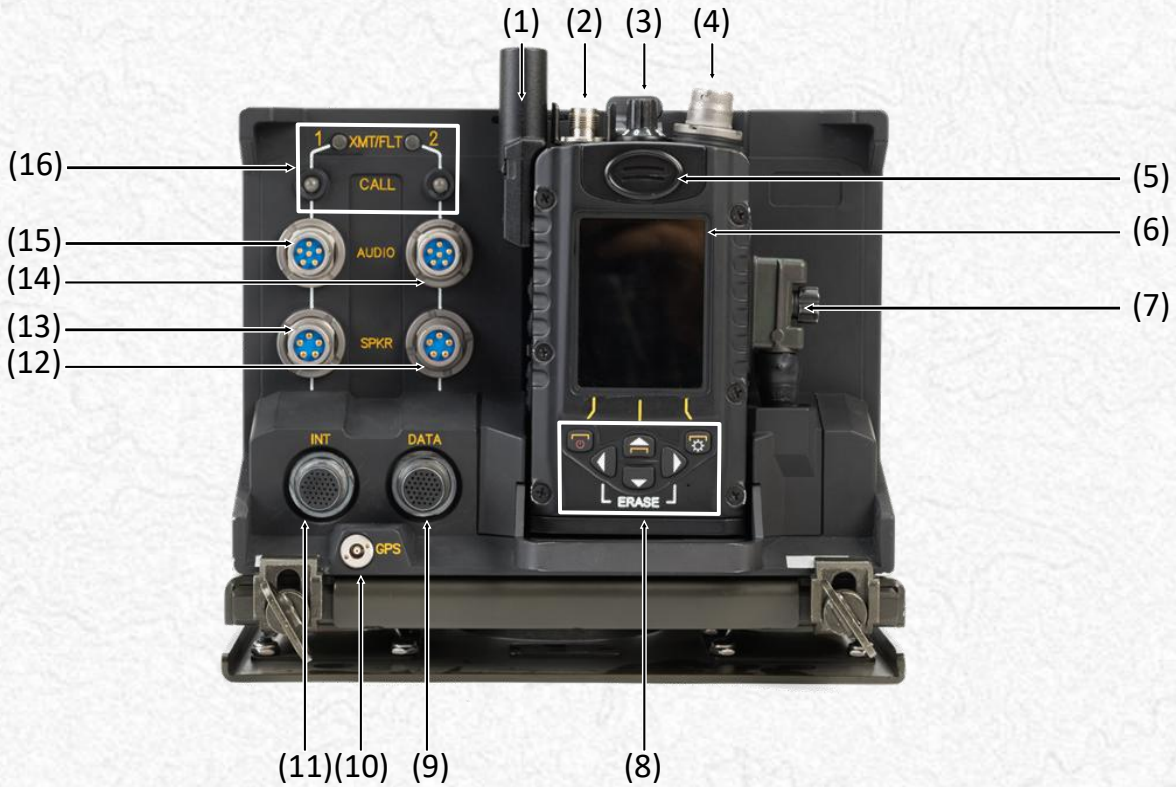


Parts & Functions Elynx MCTR 7200HH VS50-VA

- MCTR-7200HH-VS50 VA is a single channel ruggedized vehicular adapter which provides mechanical support, power and control functions for one RT-7200 unit
- MCTR-7200HH-VS50 VA includes a receiver and a RF power amplifier operating in the frequency range of 30 to 512 MHz that provides 50W

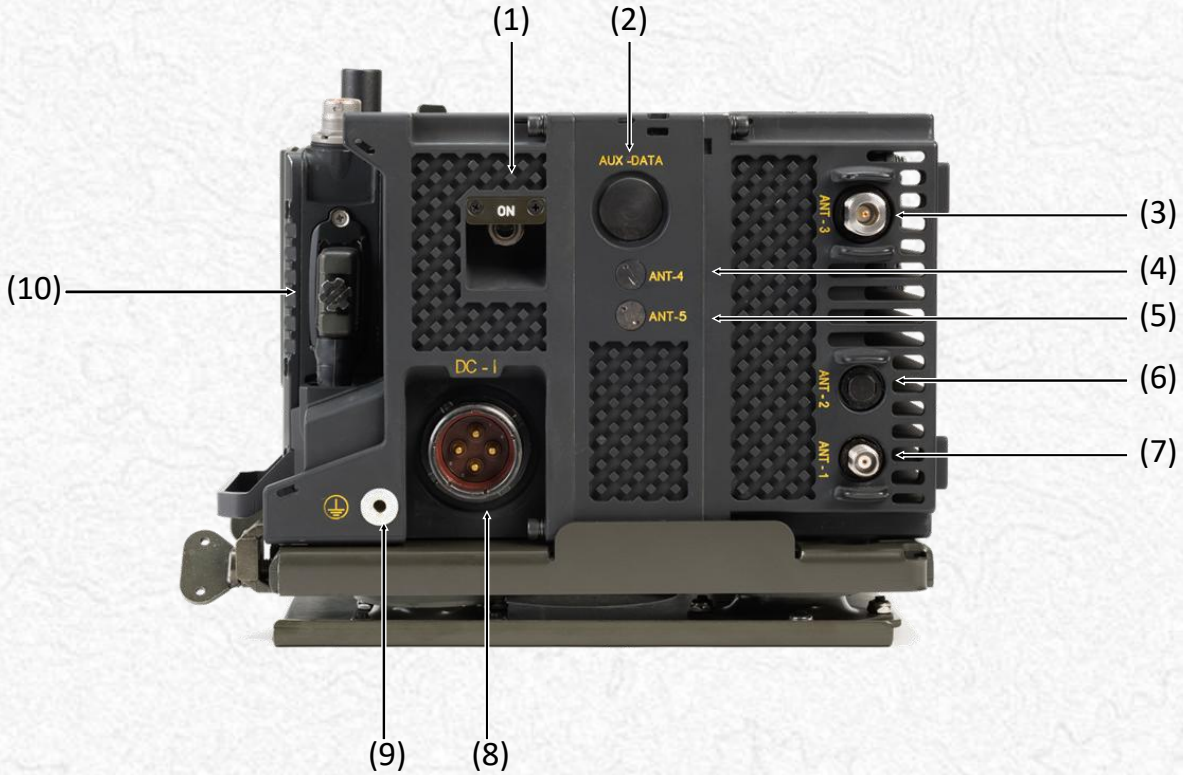


Parts & Functions Elynx MCTR 7200HH VS50-VA



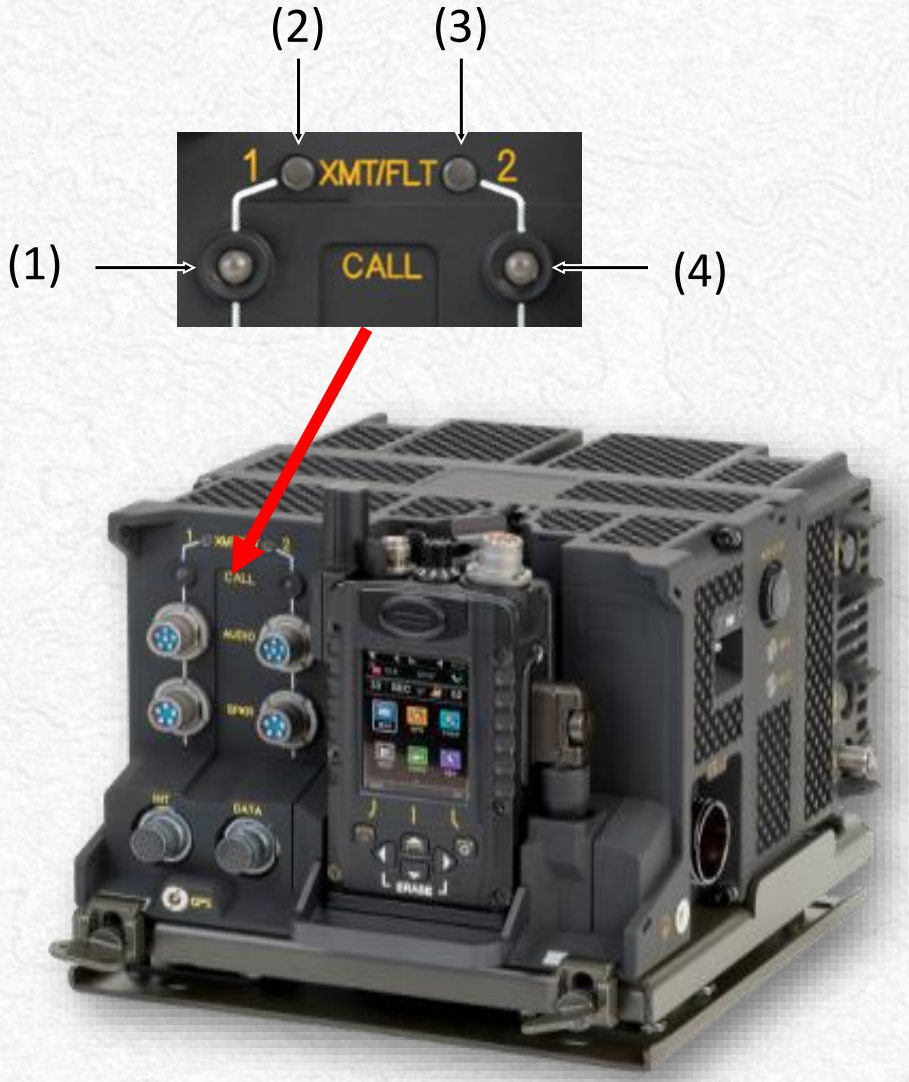
#	Item	Description
1.	GPS Antenna	GPS antenna
2.	RF Antenna Connector	RF antenna connector
3.	Channel Selector	The channel selector enables selecting between channels 0 through 9
4.	Audio Connector	Analog audio connection for handset
5.	Speaker	Internal speaker
6.	Display	2.8 inch LCD display
7.	Data Connector	22 Pin data connector
8.	Keypad	6-keys keypad used to control the GUI and turn the radio ON/OFF
9.	Data Connector (VA)	Multiuse data connector for external host
10.	GPS Connector	GPS antenna connector
11.	INT Connector	Connection to Intercom system
12.	SPKR 1 Connector	Speaker output
13.	SPKR 2 Connector	Speaker output
14.	Audio 1 Connector	Analog audio connection for handset
15.	Audio 2 Connector	Analog audio connection for handset
16.	Indication lights	Four indication lights (see section 4-2.2)

Parts & Functions Elynx MCTR 7200HH VS50-VA



No.	Item	Description
1.	Circuit Breaker switch	**High voltage protection circuit breaker, turns the VA ON/OFF**
2.	AUX Data Connector	**(N/A) reserved for future use**
3.	ANT-3	**N-Type connection to optional vehicular L-Band antenna.**
4.	ANT-4	**(N/A) reserved for future use**
5.	ANT-5	**(N/A) reserved for future use**
6.	ANT-2	**(N/A) reserved for future use**
7.	ANT-1	**TNC Antenna Connector.**
8.	DC IN Connector	**Connection to 28 VDC supply**
9.	GND Connector	**Connection to Ground Strap**
10.	Data Connector	**22 Pin data connector**

Parts & Functions Elynx MCTR 7200HH VS50-VA

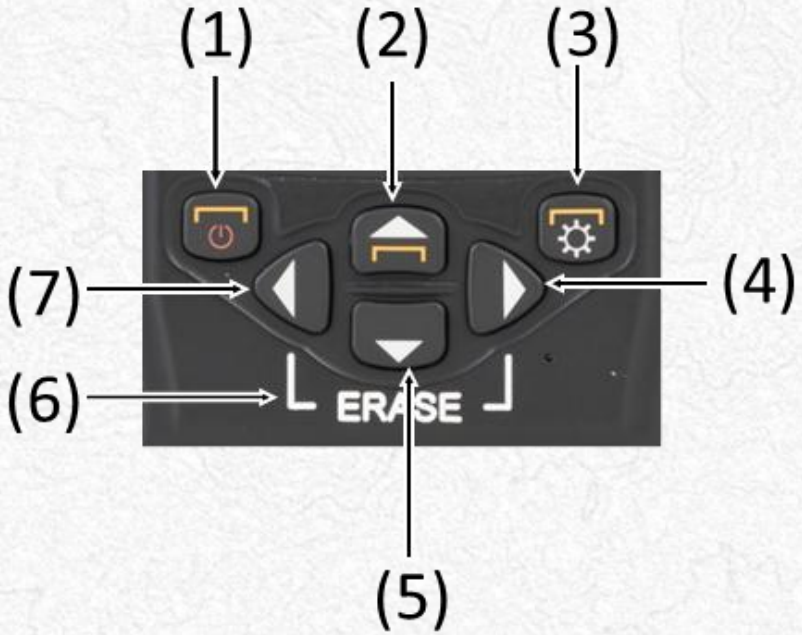


Light	Status	Color	Indication
CALL	On	Orange	The radio is receiving voice (Only when Squelch is detected)
	Off	-	The radio is not receiving voice
XMT/ FLT	On	Green	The radio is transmitting voice (The light indication duration shall be in accordance with the voice PTT duration)
		Red	Technical fault (Built-In Test error)
	Flashing	Red	Operational fault (e.g. RF antenna VSWR fault, or pressed PTT while in RCV ONLY mode)
	Off	-	The radio is not transmitting voice and there are no faults
All	On	Orange Green Red	PBIT and IBIT are in progress: Orange + Green for 2 seconds, then Orange + Red for completion of tests (If error is detected the Red indication shall remain on)

No.	Item	Description
1.	CALL-1	Indication for voice reception in Transceiver 1
2.	XMT/FLT-1	Indication for transmission/fault in Transceiver 1
3.	XMT/FLT-2	Indication for transmission/fault in Transceiver 2 (N/A)
4.	CALL-2	Indication for voice reception in Transceiver 2 (N/A)

Parts & Functions Elynx MCTR 7200HH VS50-VA

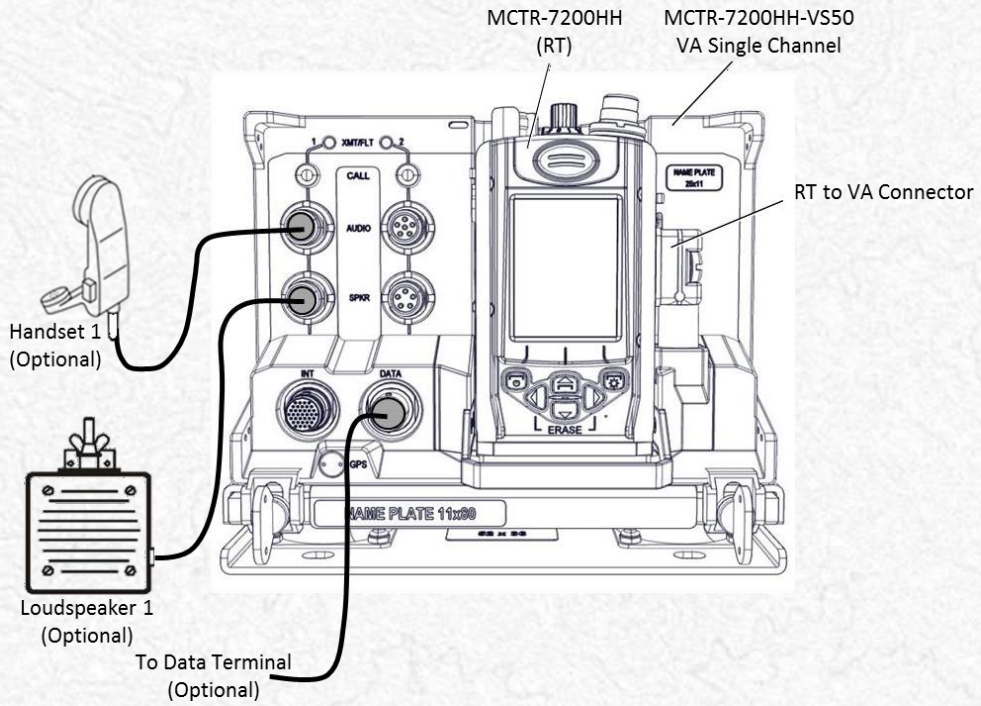
- The operation of the radio is performed using a small number of controls, including an Six-key keypad

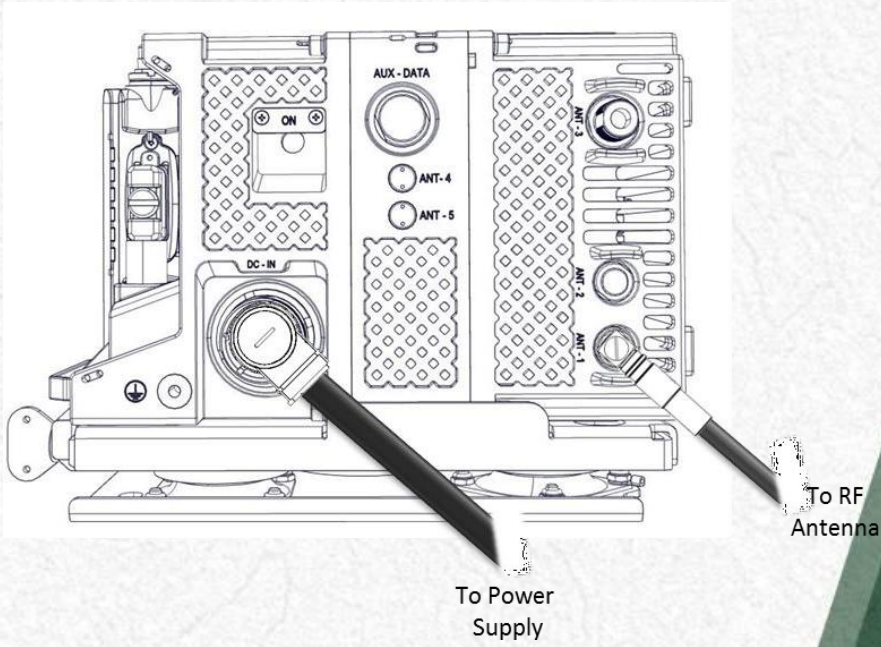


No.	Item	Description
1.	Soft key	**Used as a soft key for various application options**
2.	Up navigation key	**Up navigation key, soft key for various application options. An elongated press toggles between three screen light modes: Light ON/Night Vision Goggles mode/Light OFF.**
3.	Soft key	**Used as a soft key for various application options**
4.	Right navigation key	**Right navigation key**
5.	Down navigation key	**Down navigation key**
6.	Emergency Erase	**Pressing left (7) and right (4) navigation keys together for 3 seconds performs emergency erase (see section 4-2.1.1)**
7.	Left navigation key	**Left navigation key**

Parts & Functions Elynx MCTR 7200HH VS50-VA

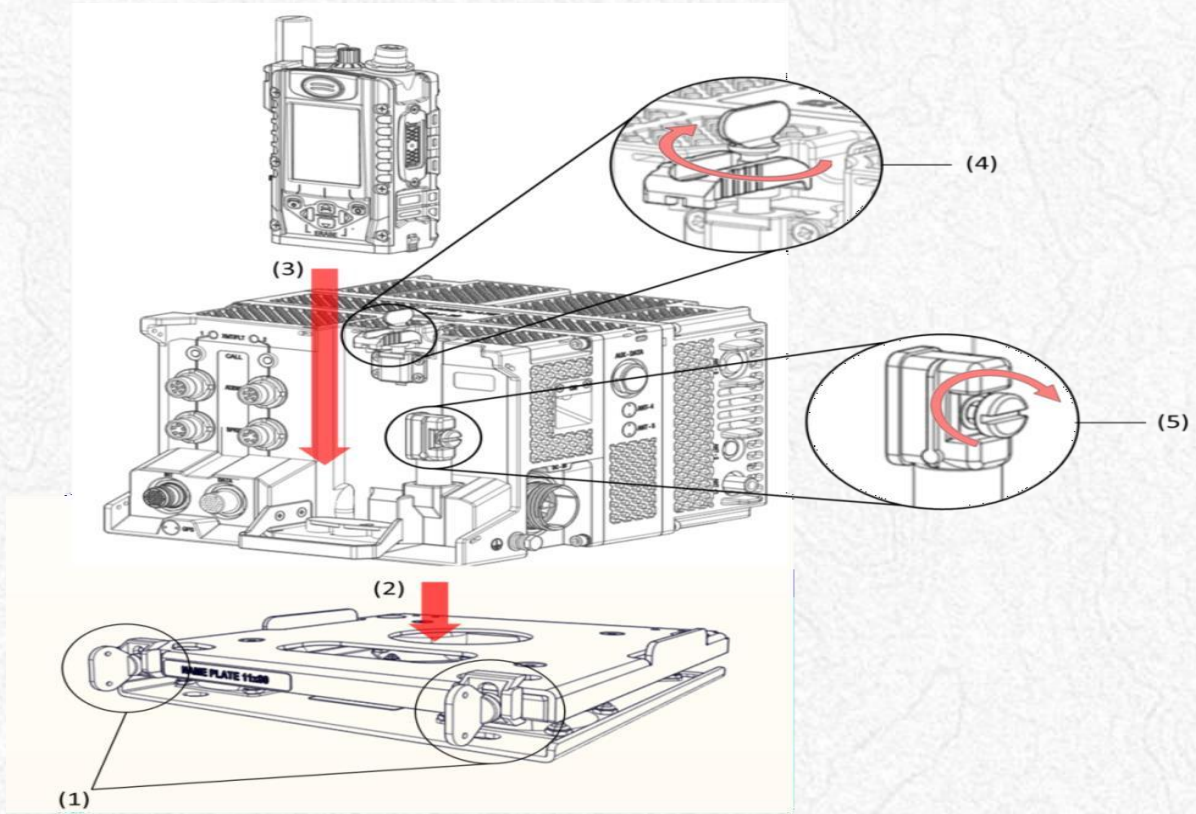
- E-LynX – HH – VS50





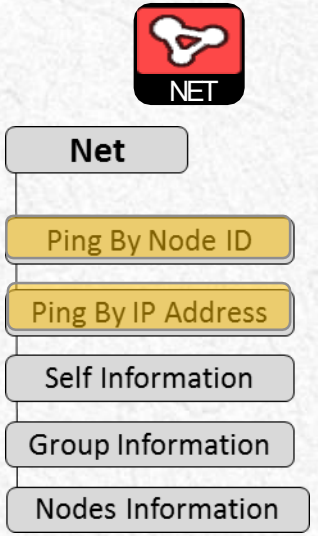
Parts & Functions Elynx MCTR 7200HH VS50-VA

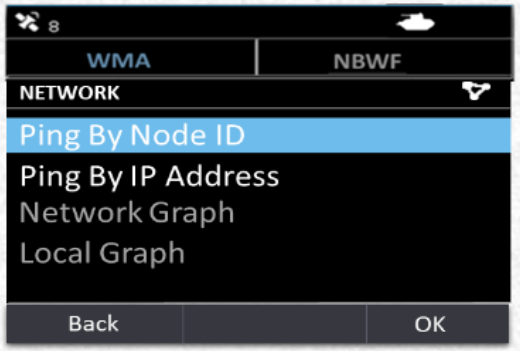
- E-LynX – HH – VS50

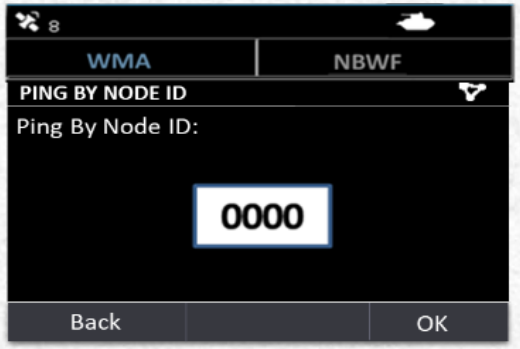


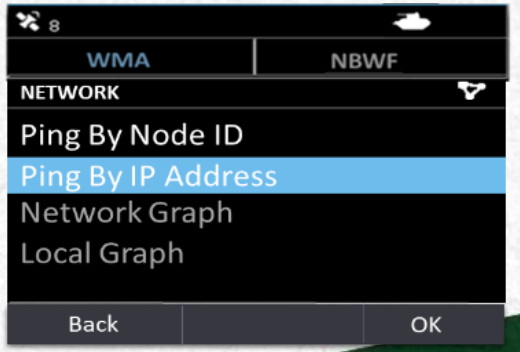
MANET Network

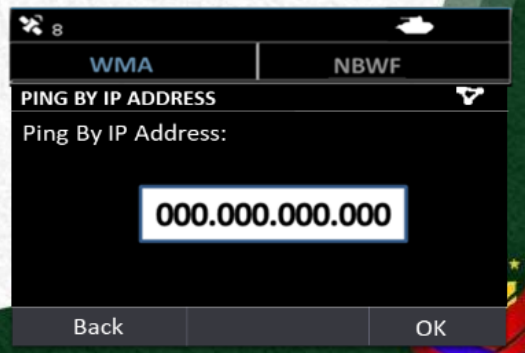
- The Network menu allows the technician to check information of the radio such as IP, Node ID , Nodes information etc.
- with another radio using pre configured unique ID.
- IP Ping- The technician can check communication with another radio or with an IP Host.
- Under Operator PSWD Menu Ping suppose to be enable





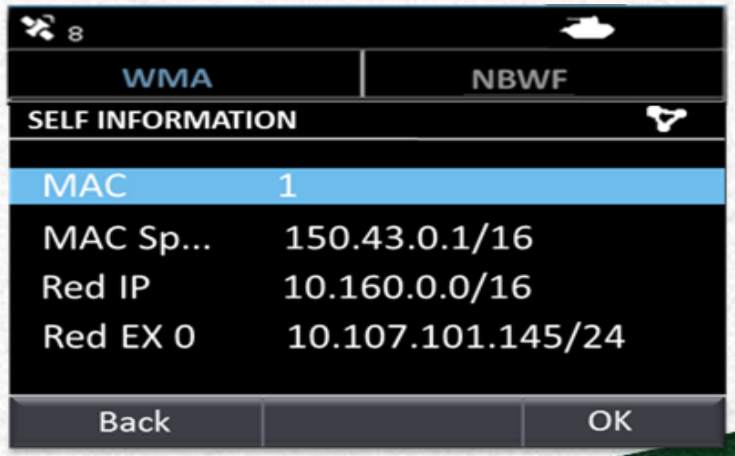


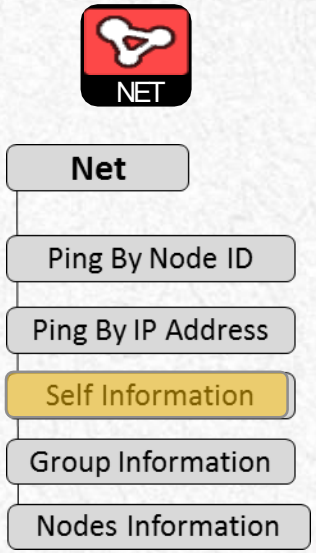




MANET Network

- The Self Information sub menu is designated for technician use and provides the following details:
  - MAC – The Node ID set by the operator in order to identify each radio.
  - MAC SPEC – A technician address that can be used to access the radio from a computer terminal.
  - Red IP – Management IP used for management purposes.
  - Red EX 0 – The external IP interface of the device.

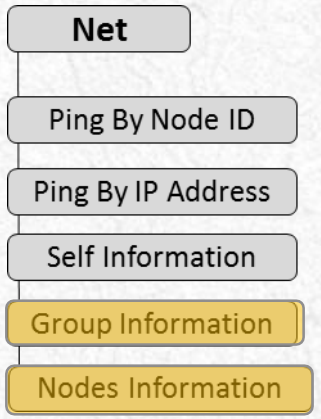


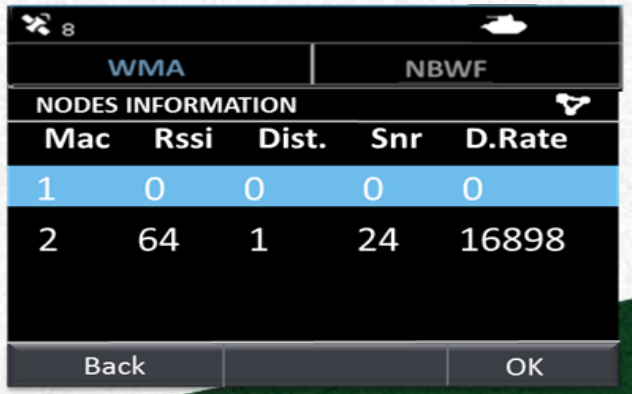


MANET Network

- Groups Information - Display information about the group voices, Such as group voice name that will be displayed in the Main menu.
- Nodes Information - Display information about the Connection of the nodes in the network.







- RSSI - Received Signal Strength
- Dist - Distance between the remote node to the local node.
- SNR - Signal to Noise Ratio, measures the received signal quality between the remote node and the local node.
- D.Rate – Data rate received from the remote node (this parameter
- is available WMA only).

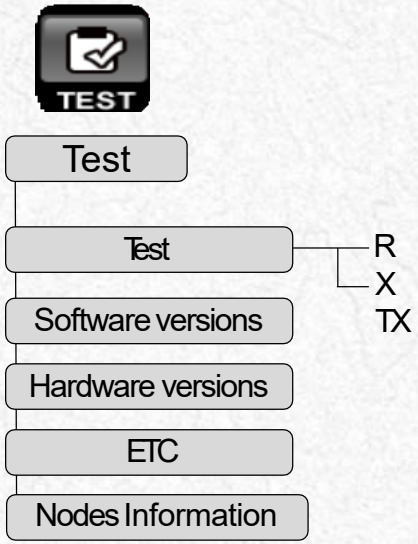



Back In 5 Minutes

MANET Network

Test

- Enable testing the RX and TX paths (including VA)
- View the software and hardware versions (for Technician level)
- View the network performance
- View the elapsed time counter



MANET Network

E-LynX™ Networking the Battlefield

Password (PSWD) Menu







MANET Network

E-LynX™ Networking the Battlefield

- Used to provide access to protected configuration and functions.

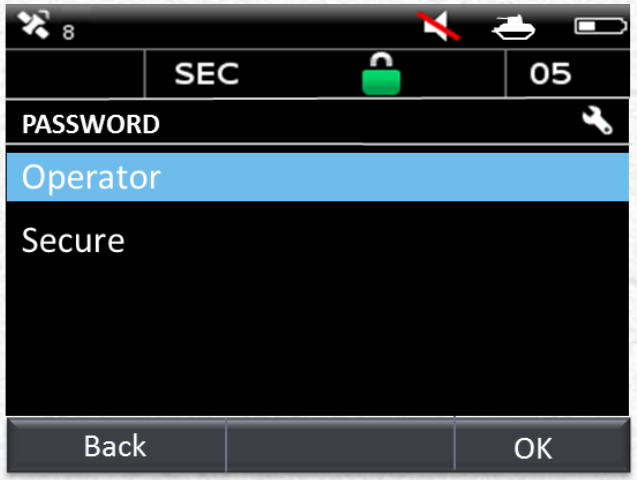


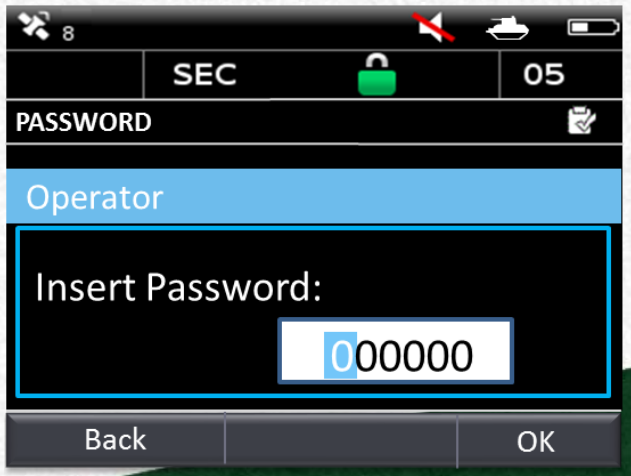
- The PSWD menu has levels, each protected by a separate password:
  - Operator level – Includes the loading of operational parameters and the selection of various global radio set parameters.
  - Secure level – Includes the loading of software updates.

MANET Network

E-LynX™ Networking the Battlefield

- To enter operator menu:
  - Navigate to the Password menu, scroll to the operator submenu and press OK.
  - Enter the Operator Password and press OK to confirm (if the password is wrong, you will see “Wrong password”, press OK and try again).





MANET Network

E-LynX™ Networking the Battlefield

Tone Settings

- The RT can modulate 150Hz tone together with the audio signal and through the PSWD menu the operator can turn it ON or OFF.



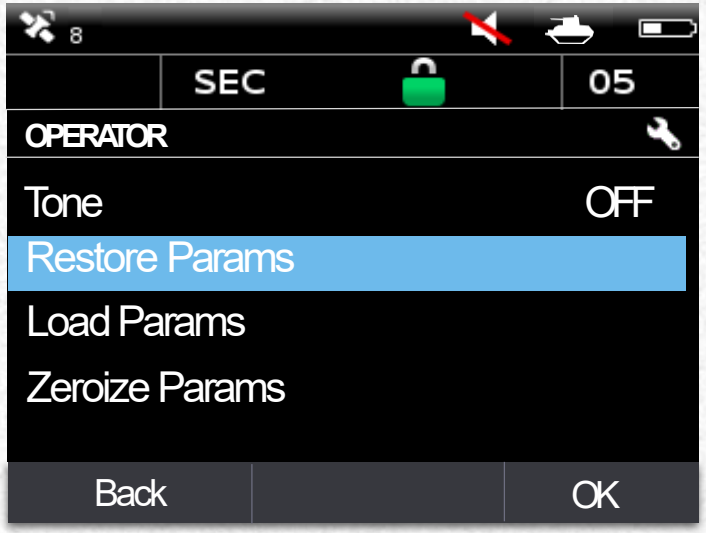


MANET Network

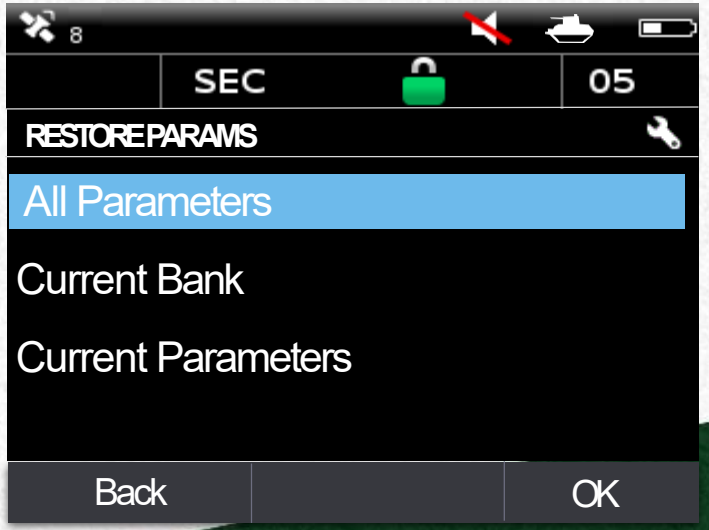
E-LynX™ Networking the Battlefield

- **Restore Parameters**
- The Restore Parameters option enables to re-load the operational parameters, which was last loaded to the device.







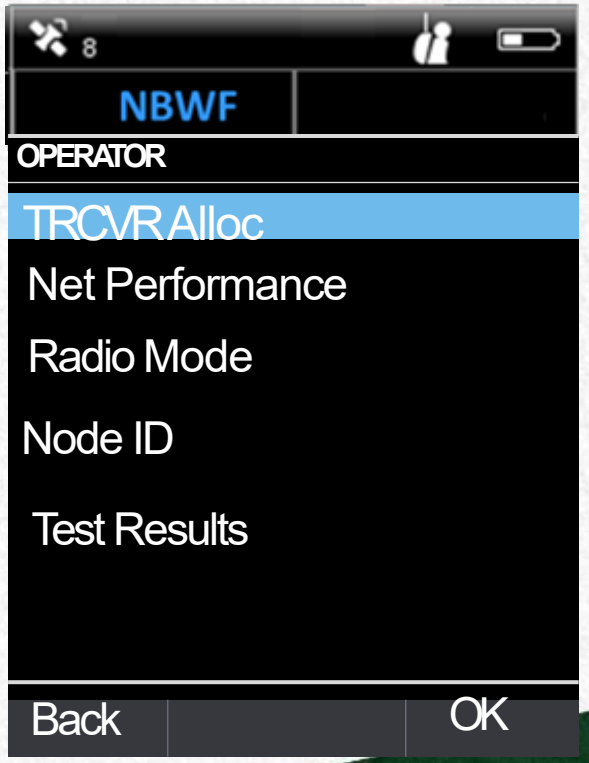


MANET Network

E-LynX™ Networking the Battlefield

Transceiver Allocation

- The Transceivers Allocation sub-menu enables selecting the waveforms to activate in each of the transceivers.



MANET Network

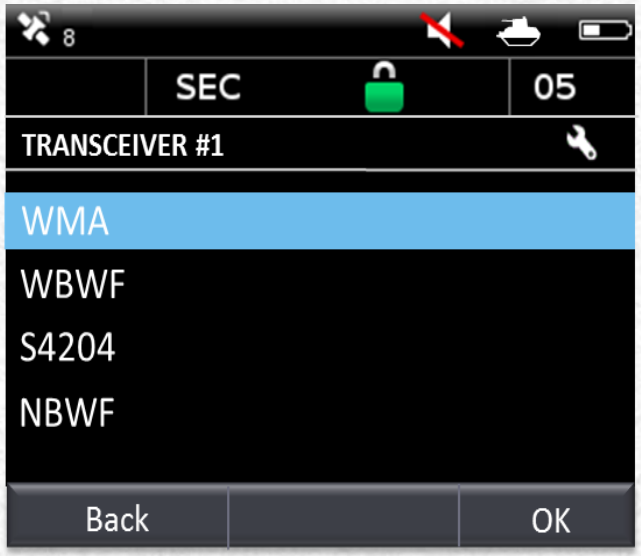
E-LynX™ Networking the Battlefield

Transceiver Allocation

PSWD

- The Transceivers Allocation sub-menu enables selecting the waveforms to activate in each of the transceivers.





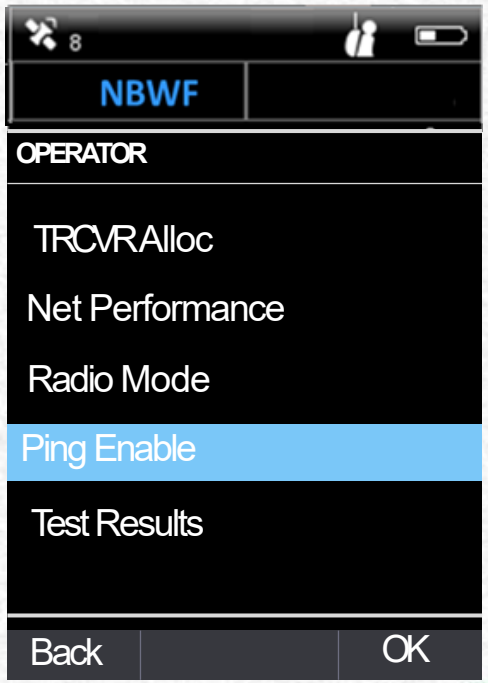


MANET Network

E-LynX™ Networking the Battlefield

Ping Enable

- Every time we turn on/off the device, we need to re-enable the option to Ping (to the radio). By activating it, we can check and ping the E-LynX radio via the PC or another E-Lynx radio



MANET Network

E-LynX™ Networking the Battlefield

- Net Performance - Option for the user to check the quality of the network by sending/checking data packets.
- Size - The size of the packets to send
- Interval - Pause time between each packet sending.
- Dest - The remote ID that the packet is sent to.
- Send-
  - Sent – How many packets were sent to the destination
  - Recv – How many packets received in the destination
- Stop Test
- Zeroize Stats - Zeroize all stats that were configured in the Net performance sub menu.



MANET Network

E-LynX™ Networking the Battlefield



Operational Mode

- The operation mode option enables selection between MTP and OTP modes:
  - OTP – One Time Programmable – Used as maintenance mode, for recovering in a case of technical fault
  - MTP – Multi Time Programmable – Used to return to normal operation mode.

NODE ID

- The node ID is a unique configuration parameter given to each transceiver.

MANET Network Integrator

H.Q. C.P. & Field Units



H.Q.

BFT / Data / Video / Voice
Remote - Services

WAN



C.P.

BFT / Data / Video / Voice
Legacy - E-LynX



Legacy
Audio
Radio

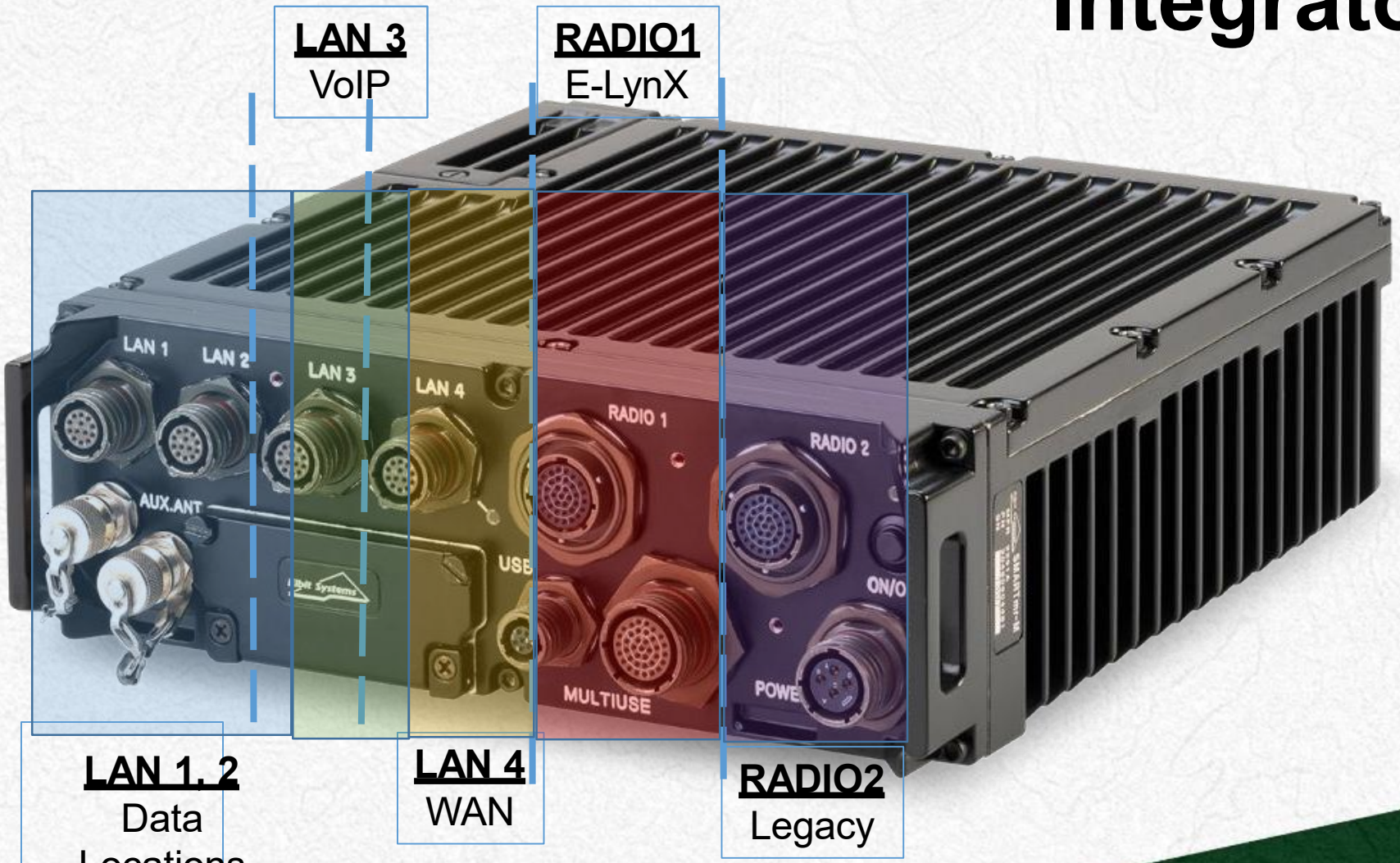


- Location
- Voice
- Data
- Video



Field Unit

MANET Network Integrator



- Best practices
- Deny traffic “leaking”
- Security
- Reduce traffic

MANET Network

Integrator



H.Q.
10.10.1.0/24
30.30.30.0/24



C.P.
10.10.2.0/24
10.107.101.0/24

- Dynamic Routing

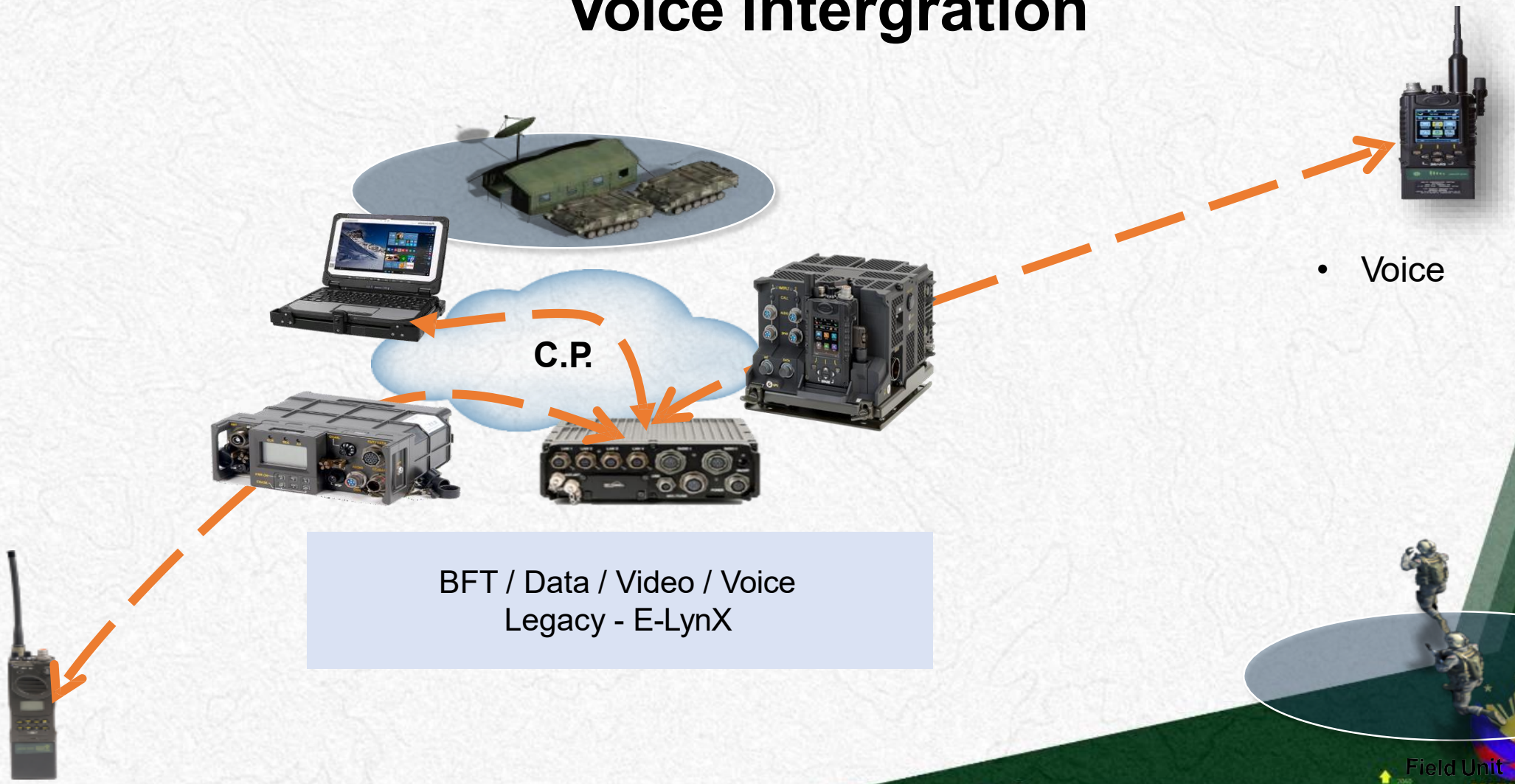
MANET Network

DATA, Positions, Video intergration



MANET Network

Voice intergration



BFT / Data / Video / Voice
Legacy - E-LynX

MANET Network

Command Post (1)







E-LynX Radio:
Node-ID 1
10.107.101.145

Integrator

Laptop CMS & Video:
10.107.101.155

Laptop iTalk:
10.10.2.1

Legacy Radio







MANET Network

Field Unit (2)



Field Unit



E-LynX Radio:
Node-ID 2
10.107.102.145

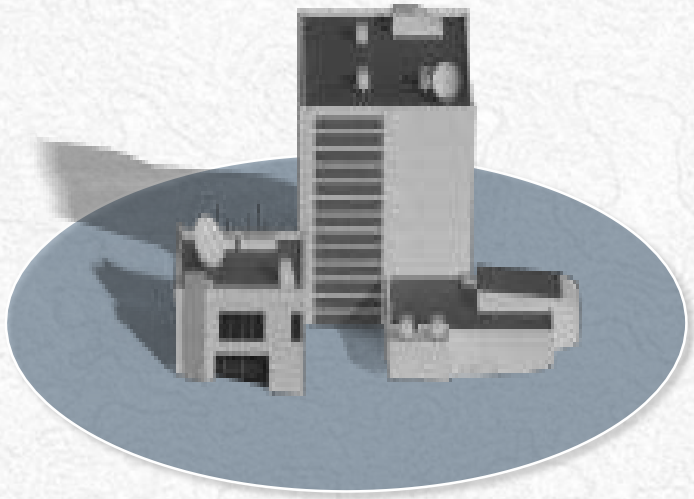
CMS Laptop

Or

IP-CAM

MANET Network

Headquarters





Integrator

Laptop

Video interface:
30.30.30.30

iTalk interface:
10.10.1.1



SUMMARY

- ✓ Introduction
- ✓ Preventive Maintenance
- ✓ Main Features and Capabilities
- ✓ Parts and Functions of HH Radio
- ✓ Parts and Functions of VS50-VA Vehicular Radio
- ✓ Network Menu
- ✓ Password Menu
- ✓ Integrator

LEARNING OUTCOME

By this time, the students will be able to demonstrate proficiency in planning and implementing MANET C4ISTAR systems to support military operations effectively in IAW PAM 6 00.

FEEDBACK

- Congratulate the learners for performing well
- Mention good performance
- Areas for improvement



FUTURE

C4ISTAR Sensors Integration




THANK YOU!
HAVE A NICE DAY
180921 22H
BEAT BOUTIQUE